When mutualism weakens: stress alters endophyte benefits in cool-season grasses

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Abstract

Benefits of plant-microbe symbioses are predicted to increase under environmental stress, but the relative roles of abiotic and biotic factors as modifiers of symbiotic interaction outcomes remain unclear. We studied how the independent and combined effects of aridity and herbivory modify the demographic effects of seed-transmitted fungal endophytes on three native grass host species using geographically distributed field experiments along a precipitation gradient, crossed with herbivore exclusion. Host demographic performance declined under lower precipitation, with endophyte-symbiotic plants having nearly twice the mortality risk of non-symbiotic plants, and spikelet production was frequently reduced under such stressful conditions[1]. Herbivory had only occasional effects on plant–microbe symbioses.[2] In contrast, abiotic stress emerged as the primary driver of symbiosis outcomes, but in the opposite direction than predicted by theory and previous studies, with symbioses becoming more beneficial in more benign abiotic environments. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

[1] *Unclear whether this is a statement about symbiont effects, or abiotic effects, or both.*

[2] *Can you add a little more detail to this statement? It's pretty vague.*

Introduction

Symbiotic mutualisms are widespread and ecologically important, occurring across plant and animal hosts, and often mediating key ecological functions such as stress tolerance and defense against natural enemies. These interactions are famously context-dependent, with the direction and strength of outcomes shaped by the environment in which they occur (Bronstein, 1994; Hoang et al., 2024; Hoeksema and Bruna, 2015). The stress-gradient hypothesis (SGH) is a classic and influential framework for predicting the context-dependency of interaction outcomes, predicting that positive interactions increase under stressful conditions (Bertness and Callaway, 1994; Holmgren and Scheffer, 2010; Louthan et al., 2015). For example, microbial symbionts that benefit host plants under drought stress may become neutral or even parasitic under more benign conditions (Giauque et al., 2019).

As symbiotic interactions shift along stress gradients, their effects are predicted to become increasingly beneficial under biotic and abiotic stress, enhancing host resilience to environmental challenges. Under biotic stress, such as attack by natural enemies, symbionts can mediate defensive traits that reduce enemy performance, survival, or reproduction across diverse taxa (Atala et al., 2022; Bastias et al., 2017; Vega, 2008; **?**; **?**). Similarly, under abiotic stress, including drought, nutrient limitation, and thermal extremes, symbiotic partners often enhance host tolerance through improved resource acquisition, physiological regulation, and stress-responsive metabolism (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). In plants specifically, microbial symbionts such as mycorrhizae, rhizobia, and endophytic bacteria and fungi can increase water and nutrient uptake, promote nitrogen fixation, and stimulate antioxidant production, allowing hosts to maintain growth and function under harsh conditions (Amijee et al., 1989). Many symbionts also stimulate the production of antioxidant enzymes, reducing oxidative damage caused by stressors such as drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023; Rudgers et al., 2020; **?**).

While stress is hypothesized to strengthen the benefits of symbiotic microbes, growing evidence suggests that it could also elevate costs of symbiosis. Maintaining symbionts requires that hosts allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of a host's photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). For example, under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to endophyte-free plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002). Thus, although stress is often expected to strengthen mutualistic outcomes, mounting evidence indicates that it can also intensify the demographic costs of hosting symbionts, raising fundamental questions about when symbioses enhance versus hinder host performance.

Despite growing interest in the role of symbiosis in shaping host demography, few studies have tested how abiotic and biotic stressors synergistically influence the demographic effects of symbionts. Furthermore, it remains unclear which of these stressors is more critical in eliciting the benefits and/or exacerbating costs of symbionts. Most work has been conducted in controlled. emphAmmophila breviligulata, drought did not intensify the effects of herbivory on growth, herbivory alone reduced performance, but endophyte-infected plants were more resistant to both stressors (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function (Reynolds et al., 2003). Understanding how abiotic stress, biotic stress, and microbial symbiosis interact to influence plant performance requires experiments that jointly vary all three factors. To our knowledge, such fully factorial studies remain rare outside controlled settings. Here, we address this gap by examining the effects of biotic and abiotic stress on the demographic effects of endophyte symbiosis on three grass host species spanning a natural range-core/range-edge precipitation gradient.

Symbioses between cool-season grasses and seed-transmitted fungal endophytes (*Epichloë* spp.) provide a powerful experimental model for studying how abiotic and biotic elements of ecological context influence host–symbiont interactions (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Because vertical transmission links host and symbiont fitness, it is expected to favor the evolution of mutualistic interactions (Gundel et al., 2011, 2024). Environmental conditions, including long-term variation in precipitation and temperature, strongly influence both the prevalence of *Epichloë* infection and the benefits conferred to hosts (Fowler et al., 2025; Rudgers et al., 2016). For example, grasses hosting endophytes often outperform endophyte-free individuals under water limitation, though the magnitude of these benefits depends on both host and symbiont identity (Decunta et al., 2021). A key benefit of *Epichloë* symbiosis is the production of fungal alkaloids that deter or reduce herbivore performance, thereby mediating plant–herbivore interactions (Clay et al., 2005;

Saikkonen et al., 2010). Hosting endophytes can also entail costs, including carbon allocation to sustain the fungus or risks associated with imperfect transmission. Finally, *Epichloë* endophyte effects can differ across host vital rates, which makes it important to consider multiple vital rates in assessing costs and benefits: symbiosis may not improve survival under stress, but can still enhance growth and reproduction (Rudgers et al., 2012; Yule et al., 2013).

We established experimental gardens of three species of native cool-season grasses across an aridity gradient in the south-central United States to investigate how endophyte symbiosis affects multiple host vital rates under varying abiotic and biotic stress. This gradient, from mesic conditions in the east to increasingly arid conditions in the west, provides a biologically realistic stress gradient, and coincides with the westernmost range limits of many eastern North American grass species. At each of seven sites along the gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus exclusion) to test its interaction with abiotic stress. Across these abiotic and biotic contexts, we quantified effects of symbiosis by comparing the demographic performance of endophyte-symbiotic (E+) and endophyte-free (E-) individuals. We asked whether endophyte symbiosis is beneficial for host grasses, whether symbiont benefits are amplified under exposure to abiotic or biotic stressors, as predicted by the SGH, and how these two forms of stress combine to influence the demographic outcomes of endophyte symbiosis. Specifically, we tested the following predictions (Fig. 1):

1. Endophyte symbiosis will generally be beneficial for the demography of these native grass hosts.
2. Because endophytes can improve drought tolerance (e.g., by enhancing water use efficiency or root growth), they will be most beneficial in arid environments, and neutral or costly in mesic environments.
3. Because endophytes produce defensive compounds (e.g., alkaloids), they will be most beneficial under herbivore exposure, and neutral or costly under herbivore exclusion.
4. The combined stresses of herbivory and low precipitation will amplify the physiological demands on the host plant, such that costs of maintaining endophyte associations will dampen net benefits relative to each stressor in isolation.

[3]

[3] *One could predict the opposite, but there is limited evidence for "synergistic benefits" of multiple stressors in plant–microbe interactions, especially for endophytes. My logic follows Johnson et al. (1997) in New Phytologist, which shows that symbiotic associations can shift from mutualistic to parasitic when environmental conditions increase the costs of symbiosis relative to benefits (Johnson et al. 1997).*

Materials and methods

Study system

Our study focused on three cool-season perennial grass species native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a short-lived perennial that germinates from autumn to late winter and, in our Texas study region, typically flowers between April and June. *E. virginicus* is a larger, longer-lived species that flowers throughout spring and summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers in late spring. *Agrostis hyemalis* hosts the endophyte *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts *Epichloë PauTG-1* or *E. typhina subsp. poae* (Gundel et al., 2020). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds and are additionally capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). In our study, stromata were never observed on *E. virginicus* or *Poa autumnalis* and very rarely observed on *A. hyemalis* (¡5% of individuals), suggesting low levels of horizontal transmission. Herbivory in this system is primarily from deer, generalist insects, and small mammalian grazers, which can influence both plant performance and endophyte-mediated defenses.

Source populations and symbiont removal

[4] Plants used in our experiment were derived from natural (source) populations of each species (Fig. 2, Table S1). Prevalence of fungal endophytes in these natural populations ranged from XX% to XX% endophyte-positive hosts (NEW SUPP TABLE). [5] Seeds from these source populations (from ca. 30 plants per population) were either heat-treated or left untreated, intended to produce symbiont-free (E^{-}) and symbiotic (E^{+}) plants from the same genetic lineages. For *Elymus virginicus* and *Poa autumnalis*, heat-treated seeds were placed in a drying over at 60°C for approximately five days. For *Agrostis hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 minutes. [6] Where possible, endophyte-removal lineages were reared through at least one generation before use in experiments, to minimize the direct

[4] *I am planning to work on the supp data for this section but for now I want to keep moving, so just editing the text.*

[5] *I am not sure about this.*

[6] *We should talk about this. You need more detail here – I have tried adding some but there is more to do. They were all treated but not all treatments were successful, which we need to report. And there were a few different methods used on different species. This should all go in the supplement. It is also not true that heat treatment has no effect on seed viability, which is why we tried to use seeds that were multiple generations removed from the heat treatment.*

effects of heat treatment (Supp Table). All seeds, both heat-treated and non-heat-treated, were surface-sterilized with 10% bleach solution then germinated on agar plates. Seedlings were transferred to potting soil and grown in the Rice University greenhouse. Once these plants were sufficiently large in size they were scored for their endophyte status and vegetatively propagated, with clonal identities tracked across individuals. In total, we used 3 source populations and XX unique genotypes for *A. hyemalis*, XX source populations and XX genotypes for *E. virginicus*, and XX source populations and XX genotypes *P. autumnalis*. After vegetative propagation, the experiment included 840, 840, and 720 clones in total across all populations.

Before transfer to the field experiment, we confirmed endophyte status (E^+ or E^-) of all seedlings using either microscopy or an immunoblot assay; realized numbers of E^+ and E^- individuals per species and population are provided in Supplementary Table S3. We assessed endophyte status using two methods. First, for microscopy, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae (Jackson and Johnson-Cicalese, 1988). Second, we used an immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) that target proteins of *Epichloë* spp. to visually indicate presence or absence (Koh et al., 2006). Both approaches have proven reliable in previous studies and, in cases where both were performed ($N = XX$), yielded similar results in our study (stats). For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assume that its status applies equally to all clones of a given source genotype.

Common garden experimental design and climate data

To examine the effects of endophyte symbiosis across an abiotic stress gradient, we established common garden experiments with E+ and E- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S2). From west to east, mean annual precipitation increases more than three-fold across these sites while mean annual temperature is relatively consistent, based on climate data collected during the same period as our demographic surveys (Fig. S1). For this reason, we focus on precipitation as the primary abiotic factor that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring exposure to realistic climatic and biotic conditions (Fig. 2A,B, C).

To characterize climate at these sites during the study period (2023–2025), we collected monthly temperature and precipitation data from PRISM at 800 m resolution (Hart and Bell, 2015). For each site, we calculated mean temperature (°C) and cumulative precipitation (mm) over the corresponding census year, defined as the period from transplanting to the demographic census. During our study period, realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites (Fig. 2D).

At each of seven sites along the gradient, we established 24 square plots (1.5 m $\times$ 1.5 m), eight per host species (*P. autumnalis* was planted at six sites, excluding the westernmost site (SON) since it is well outside this species' natural range: Fig. 2C). Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2E). The herbivory treatment targeted both vertebrate and invertebrate hebivores: exclusion plots had deer fencing on four sides and were sprayed with Sevin insecticide (active ingredient Zeta-Cypermethrin) 2-3 times per growing season following the manufacturer's protocol; herbivore access plots were unsprayed and fenced on two sides to control for any fence effect.

Greenhouse-raised plants were planted into the field experiment in February-March 2023. Each plot was planted with 15 individuals, a mix of E+ and E- host-plants. Clonal replicates were arranged such that all plots had the same level of genetic diversity (number of unique genotypes) for each species. For each species at each site along the precipitation gradient, we planted a total of 60 E+ and 60 E- host-plants, half of those experiencing herbivore access and half experiencing herbivore exclusion. Plots at the two westernmost site (KER and SON) were destroyed by deer and subsequently replanted in 2024 with stronger fencing.

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May and *Elymus virginicus* was censused in June of 2024, and 2025. For each individual, survival was recorded as alive or dead and, for surviving plants, size was measured as the count of tillers. Growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in years t and $t+1$. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*. In *A. hyemalis*, one spikelet yields one seed, while in *P. autumnalis* and *E. virginicus* a single spikelet can produce 2-6 seeds. We measured herbivore damage to assess the effects of herbivory treatment on realized herbivory experienced by individual plants. We defined damage as chewing damage to leaves, tillers, or inflorescences, and estimated the proportion of damaged plants by plot, site, and herbivore treatment. These data showed that plants in fenced plots experienced less herbivory than in unfenced plots (Fig. 2F,Fig. S2,Fig. S3), confirming that our experimental manipulations had the intended effects.[7]

Statistical modeling

To assess how abiotic and biotic stress modify the demographic effects of endophyte symbiosis, we fit generalized linear mixed models for each vital rate response in a hierarchical Bayesian statistical framework. All models shared the same hierarchical structure for their expected values and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, inflorescence count with a zero-inflated negative binomial distribution, and spikelet count with a negative binomial distribution. All models included species-specific fixed effects for symbiosis (E+ or E-), precipitation (cumulative rainfall over 12 months preceding the census), and herbivory (herbivore access/exlusion), including all two- and three-way interactions. We considered models with linear and quadratic forms for the influence of precipitation, to account for the possibility of non-monotonic responses. Although we evaluated quadratic models, model selection criteria (LOOCV) showed that they provided minimal improvement over monotonic models (Table S4). Therefore, all subsequent analyses used the first-order (linear) effect of precipitation, which captures the observed patterns while maintaining model simplicity. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case. [8]

For each vital rate model, the expected value for the i^{th} observation (μ_i) was given by:

$$
\begin{aligned}
\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i]precip_i + \beta_{P^2}[Sp_i]precip_i^2 + \beta_E[Sp_i]endo_i + \beta_H[Sp_i]herb_i \\
& + \beta_{EP}[Sp_i]endo_i precip_i + \beta_{HP}[Sp_i]herb_i precip_i \\
& + \beta_{HE}[Sp_i]herb_i endo_i + \beta_{HEP}[Sp_i]endo_i herb_i precip_i \\
& + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]} \\
& \qquad \phi_{\text{site}[i],Sp_i} \sim N(0, \sigma^2_{site}) \\
& \qquad \rho_{\text{plot}[i],Sp_i} \sim N(0, \sigma^2_{plot}) \\
& \qquad \omega_{\text{source}[i],Sp_i} \sim N(0, \sigma^2_{source})
\end{aligned}
\tag{1}
$$

[7] *I agree with you that Methods is a better place for this. But somewhere we should address how the magnitude of herbivory and effectiveness of exclusion varies across sites, which reflect natural background variation in the herbivore communities. Reviewers may also ask to see how herbivory differed between E+ and E- plants, which I think could be an interesting and relevant addition. You don't really analyze the herbivory data beyond showing this figure, so I think this could be something that would enrich the results, or maybe be appropriate for the supplement.*

[8] *I suggest NOT doing this, if you are saying that the results are all based on the linear model- just show the linear model, it's easier to digest.*

Here, $precip_i$ is total precipitation (mm) accumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence ($endo_i = 1$) or absence ($endo_i = 0$), and $herb_i$ indicates fencing treatment: herbivore access ($herb_i = 0$) or exclusion ($herb_i = 1$). The model includes random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity that is not explained by precipitation, $\rho_{\text{plot}[i]}$ captures plot-to-plot variation within sites, and $\omega_{\text{source}[i]}$ captures variation associated with the provenance of transplants (the latter two are unique to species and therefore not indexed by species). Random effect for site, plot, and source are host species-specific but the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species; this assumes, for example, that site-to-site heterogeneity is of a similar magnitude across species, even as the model allows for "better" and "worse" sites to differ by species. This hierarchical structure allows the model to"borrow strength" across species (Gelman and Hill, 2007), improving estimates of fixed effects while still capturing species-specific responses to environmental gradient. [9]

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024). Models were run with 4 chains and sufficient iterations to ensure convergence, with weakly informative priors specified for all fixed effects. Full details on priors and model settings are provided in the accompanying code. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using posterior predictive checks (Fig. S4).

We used posterior parameter estimates to visualize predictions for how effects of symbiosis varied across abiotic and biotic contexts. For each herbivory treatment (Access or Exclusion) at each level of precipitation, we computed posterior predictions for survival probability of E^+ and E^- hosts using the Bayesian model outputs (Eq. 1). The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits or a costs on vital rates under the given precipitation and herbivory conditions. When endophytes conferred a measurable cost,[10] we additionally quantified the effect using a ratio-based metric (E^+ / E^-), where 1 indicates no effect, <1 indicates reduced performance of E^+, and >1 indicates enhanced performance. Both Δ and ratio metrics were summarized for low-precipitation (abiotic stress) conditions, defined as the lowest precipitation quartile, and interpreted only when posterior distributions indicated

[9] *Notice my change to the random effects in the equation. I think this is the more correct way to show plot and source effects, since they are already unique to species, and this should not require any changes to the code.*

[10] *Unclear why you only did this in the case of costs*

a significant difference between E^+ and E^- (i.e., 90% credible intervals excluding 0 for Δ or 1 for ratios), ensuring that reported costs or benefits reflect biologically meaningful effects. [11]

Comparing drivers of endophyte effects on host demography

To quantify relative statistical support for abiotic and/or biotic contexts as modifiers of grass-enodphyte symbiosis, we fitted three hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production; Supplementary Methods S.1.1). All models included endophyte status as a baseline predictor but differed in whether they incorporated interactions with precipitation (Endophyte $\times$ Precip), herbivory (Endophyte $\times$ Herbivory), or both factors, including the three-way interaction (Endophyte $\times$ Precip $\times$ Herbivory).[12] We compared candidate models using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD), along with associated standard errors, were used to rank model performance. When the difference in ELPD between two models (ΔELPD) was small relative to its standard error (SE), we considered their predictive performance to be effectively similar. Accordingly, model rankings were interpreted cautiously, focusing on consistent trends across vital rates rather than on small differences between closely ranked models.[13]

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Wetter conditions increased host survival across all species and abiotic stress modified endophyte effects on survival for two of the three grasses, but in the opposite direction as predicted: grass-*Epichloë* symbiosis generally had negative or neutral effects under low precipitation (Table S5), but benefits increased under wetter conditions (Fig. 3; Fig. S5;Table S6). Under herbivore access, *Agrostis hyemalis* and *Elymus virginicus* showed median $\Delta(E^+ - E^-)$ values from -0.07 to -0.01 at low precipitation ($\Pr(\Delta > 0) = 0.15$–0.42), increasing to 0.06–0.15 under high precipitation ($\Pr(\Delta > 0) = 0.78$–0.91). Herbivore exclusion weakened these effects:[14]

[11] *I don't understand why you use both Delta and the ratio. Don't they provide the same information? Added later: it also seems like you only focus on the delta values in the Results.*

[12] *There is a fourth option: Herbivory + Precipitation with no interaction. Did you consider that model?*

[13] *This is really nicely written.*

[14] *I don't see this in the figure. It actually looks like a stronger switch from costly to beneficial in herbivore exclusion.*

in *Agrostis*, Δ remained near zero across most precipitation levels, whereas *Elymus* shifted from strongly negative under dry conditions to positive under wetter conditions. *Poa autumnalis* showed little evidence of survival benefits under herbivore access ($\Delta = -0.05$ to 0.00; $\Pr(\Delta > 0) = 0.12$–0.53) and weakly positive effects under herbivore exclusion ($\Delta = 0.00$–0.03), with credible intervals overlapping zero. [15]

Endophyte[16] presence influenced growth in all species, depending on herbivore access and precipitation (Fig. 4;Table S7). In *A. hyemalis*, growth effects were neutral to slightly negative under dry conditions and became increasingly positive at moderate-to-high precipitation when herbivores were present ($\Delta = 0.27$–0.62; $\Pr(\Delta > 0) = 0.94$–0.99). Benefits under herbivore exclusion were smaller but consistently positive ($\Delta = 0.08$–0.41; $\Pr(\Delta > 0) = 0.71$–0.96). *E. virginicus* exhibited weakly positive growth under herbivore access (Δ increasing modestly from dry to wet; $\Pr(\Delta > 0) = 0.52$–0.96) but small and uncertain effects under exclusion. [17] In *P. autumnalis*, growth responses varied with precipitation and herbivore access: under herbivore access, Δ shifted from negative in dry conditions to positive in wet conditions, whereas under exclusion, benefits were strongest at low precipitation ($\Delta = 0.56$–0.81; $\Pr(\Delta > 0) = 0.97$–0.98) but declined and became negative at the highest precipitation levels ($\Delta = -0.41$; $\Pr(\Delta > 0) = 0.03$).[18] [19]

Effects of *Epichloë* on reproduction were trait and species-specific.[20] Inflorescence production showed little evidence of endophyte benefits in *A. hyemalis* or *E. virginicus*, with medians near zero and high uncertainty (Fig. 5;Table S8). By contrast, *P. autumnalis* exhibited strong positive effects, particularly under herbivore exclusion (Table S8). Spikelet production followed a similar pattern. *E. virginicus* and *P. autumnalis* had weak or negative endophyte effects under low precipitation, shifting to positive under wetter conditions, with benefits generally stronger when herbivores were excluded (Fig. S8; Fig. S9; Table S9). Credible intervals overlapped zero at low to moderate precipitation, indicating strong reproductive benefits were largely restricted to wetter conditions.

[15] *Survival figure comment: if you are going to visualize plot means, can you make point sizes proportional to sample sizes of the plots? I would also jitter and add transparency to the points. Values near zero and one are getting cut off- can you extend the margins a bit? The background gridlines make the figure bery busy IMO.*

[16] *The survival results started with overall response to the abiotic gradient, and I think that would be good here too.*

[17] *Unclear whether this sentence is about the overall growth trend or the endophyte effect. I think in general the results can be described more clearly in this regard.*

[18] *For herbivore access it looks like there is basically no endophyte effect, so don't try to oversell this. For both access and exclusion you are extrapolating the fitted models to Sonora, but this species was never in Sonora, so I would end this function at Kerrville.*

[19] *Comments on the growth figure: can you label the y axes as log ratio of size? The axis fonts and numbering are tiny on all the figures - would help to make this bigger. I would favor showing individual level growth rate rather than plot means. If you show means you should also show SE's and make point size proportional to sample size.*

[20] *Same comment as above: I would structure the results consistently so that you always mention how the vital rates varies across the abiotic gradient.*

Abiotic factors as the primary drivers of fitness components across the stress gradient

The outcomes of grass-endophyte symbioses, whether beneficial, costly, or neutral, were primarily associated with climatic variation rather than herbivore treatment (Table 1). Across all vital rates (survival, growth, inflorescence, and spikelet production) and host species[21], model comparisons consistently indicated that models including an *Endophyte* × *Climate* interaction had equal or greater predictive support than alternatives emphasizing herbivory. [22] Although differences in expected log predictive density (ELPD) were modest relative to their standard errors, these patterns were consistent across fitness components. For survival, the *Endophyte* × *Climate* model showed slightly higher predictive performance than the *Endophyte* × *Herbivory* model (ΔELPD $= -2.1$, SE = 2.8), whereas inclusion of the three-way *Endophyte* × *Climate* × *Herbivory* interaction reduced predictive performance (ΔELPD $= -5.0$, SE = 1.9).[23] [24]

Discussion

Drawing on three years of field experiments and Bayesian demographic models, we evaluated how biotic and abiotic stressors jointly influence the demographic effects of vertically transmitted *Epichloë* endophytes on three cool-season perennial grasses across a natural climatic gradient in the southern United States. We predicted that endophytes would enhance host fitness in arid environments and in the absence of herbivores, representing an endophyte-driven stress amelioration scenario (Fig. 1A), and that the addition of herbivory would reduce or reverse these benefits due to compounded metabolic costs (Fig. 1B). Contrary to these expectations, we found little evidence that endophytes buffered hosts against declining precipitation, even when herbivores were excluded. Instead, endophyte-mediated benefits were strongest under relatively benign, mesic conditions and weakened as abiotic stress intensified toward these host species' arid range edges, while herbivory played a secondary and inconsistent role.[25] These patterns do not support SGH-based predictions and instead suggest that increasing abiotic stress constrains both host performance and the capacity of endophytes to confer demographic benefits.

[21] *I added this - is that correct?*

[22] *I think you can phrase this more strongly. For all vital rates endo*climate was the best model, and that seems like the result to lead with.*

[23] *This text does not match the values in the table, and I am not sure which is correct. Based on the table, endo*climate is strongly favored over both other models. For the other vital rates, model rankings are closer.*

[24] *You have a Figure 6 in the manuscript that is not described in the results.*

[25] *It would be nice in this paragraph to more directly acknowledge the model selection results, which confirm climate as the key factor affecting endophyte benefits.*

Endophyte presence often enhances host performance when abiotic resources, such as water, are sufficient to support both host and fungus, a pattern widely documented across grass-*Epichloë* systems (Clay, 1990). Endophytes can increase host growth, reproduction, and survival (Rodriguez et al., 2009), with benefits typically strongest under low abiotic stress, such as wetter conditions (Clay, 1988; Gibert et al., 2012), because hosts must invest carbon to maintain the symbiosis. Endophytes may also help hosts cope with environmental stress by promoting compounds that reduce oxidative damage during drought, pathogen attack, or other stressors (White Jr and Torres, 2010). Although endophytes often confer benefits under favorable conditions, these advantages can decline as abiotic stress increases. Under low-precipitation conditions, reduced water availability may limit nutrient uptake and increase the metabolic burden of maintaining the endophyte, potentially tipping the cost-benefit balance against symbiotic plants (Ahlholm et al., 2002; Bronstein, 2001; Faeth, 2002). One possible mechanism is that severe drought can activate plant defense pathways, including elevated phytohormone production and reactive oxygen species, which may inhibit endophyte growth or reduce its functional contributions to the host (Bastías and Gundel, 2023; Eaton et al., 2011). Importantly, in our study system, mutualism weakening occurred even in the absence of herbivores, indicating that abiotic stress alone can disrupt the host-endophyte relationship. This contrasts with studies emphasizing herbivore-mediated context dependence (Afkhami and Rudgers, 2009; Bastias et al., 2017) and suggests that environmental limitations can independently generate similar outcomes.[26]

Drought simultaneously restricts water and nutrient availability while increasing metabolic demands; thus, the host's carbon budget may be a key factor shaping the net costs and benefits of symbiosis. Water-stressed plants close stomata to conserve water, reducing photosynthetic carbon gain while still expending C to maintain cellular homeostasis (Dietze et al., 2014). Under these conditions, sustaining an endophyte may represent a relatively high carbon cost. Similar to other nutritional mutualisms, such as nitrogen-fixing rhizobia, water stress could reduce C allocation to the symbiont, although microbial partners may partially offset these costs by enhancing photosynthesis or water-use efficiency (Pringle, 2016; Toby Kiers et al., 2010).[27] While carbon is a primary integrative currency, other factors including nutrient limitation and shifts in plant hormones also likely influence symbiosis outcomes. Future studies could quantify these costs by measuring nonstructural carbohydrates

[26] *This is a well-written paragraph, but it makes it sound like we already knew that endophytes were more beneficial in benign environments, in which case the predictions we make in the Intro are completely wrong! I think we need to fine-tune this so that we can first present the results as surprising (see for example the Decunta et al. review showing endophyte drought benefits), but then pointing to literature that could help explain the surprise.*

[27] *This is really interesting!*

(NSCs) across tissues and over time along precipitation gradients[28], providing a clearer picture of how carbon limitation, in combination with other stressors, shapes the net outcome of endophyte-host interactions. Such approaches would improve understanding of the energetic balance of these interactions under varying abiotic conditions and help predict how environmental change may shift mutualism outcomes, with implications for both ecological theory and the management of plant-microbe systems.

Across the precipitation gradient, endophyte presence had little effect on overall inflorescence production, whereas spikelet production at the driest sites tended to be lower in endophyte-symbiotic plants. Although these effects were modest and uncertain, they suggest that drought-driven resource limitation may disproportionately constrain specific reproductive components. In low-resource environments, plants often delay reproduction, grow to a larger size before allocating energy to flowers and seeds, and may ultimately produce fewer reproductive structures (Wilson and Thompson, 1989). Consistent with this, spikelet production was more sensitive to resource limitation than inflorescence initiation, whereas overall vegetative growth remained unaffected by endophyte presence. This indicates that late-stage reproductive investment is constrained under high abiotic stress, even when vegetative growth is maintained. Such trait-specific sensitivity highlights the importance of examining multiple fitness components when evaluating symbiotic costs and benefits and aligns with demographic compensation, whereby reductions in one component may be partially offset by stability in others (Cheplick and Faeth, 2009). [29]

Abiotic stress was the primary driver of endophyte-mediated effects across species and fitness components, while herbivory played a secondary role that depended on site-specific herbivore pressure. Water limitation strongly influenced host demography in our system, with drier sites constraining survival, growth, and fertility. Contrary to predictions of the stress gradient hypothesis (SGH), which posits that biotic interactions should become more positive under stressful conditions (Lortie and Callaway, 2006; Louthan et al., 2015), endophyte-mediated benefits were reduced under high abiotic stress. Most empirical tests of SGH have focused on plant–plant interactions (Bertness and Callaway, 1994), and while some recent syntheses suggest microbial interactions may show stronger support for SGH (Hammarlund and Harcombe, 2019; Piccardi et al., 2019), our results indicate that host-associated plant microbe mutualisms may not conform to this pattern.[30] Unlike neighboring plants, which can reduce environmental

[28] *We should do this! Let's discuss.*

[29] *I found this paragraph challenging because I am not sure how much you are talking about direct effects of the precipitation gradient on plant investment, versus effects of the precipitation gradient on changes in the symbiosis. Different sentences mention these at different points.*

[30] *Michelle Afkhami's mitigation–exacerbation paper is very relevant and should be cited here. We should also look at what recent papers have cited that paper.*

stress for themselves and others for example, by shading soil, retaining moisture, or improving nutrient availability endophytes live entirely within their host and cannot directly modify the external environment. As a result, the benefits they provide depend on how well the host is coping with stress. When survival, growth, or reproduction is strongly limited by drought, the endophyte's capacity to enhance host fitness may be reduced. More broadly, our study suggests that outcomes of endophyte-mediated mutualisms are shaped not just by the presence of stress but by how stress constrains host performance, with important implications for how symbiotic responses may change under increasing drought associated with climate change[31].

[31] *I think it would be good if we could expand the discussion of climate change. There is a lot of literature on microbial mutualisms and climate change, so we should reference more of that, and tie in our results. We may also want to reference Josh's GCB paper.*

References

Afkhami, M. E. and Rudgers, J. A. (2008). Symbiosis lost: imperfect vertical transmission of fungal endophytes in grasses. *The American Naturalist*, 172(3):405–416.

Afkhami, M. E. and Rudgers, J. A. (2009). Endophyte-mediated resistance to herbivores depends on herbivore identity in the wild grass festuca subverticillata. *Environmental Entomology*, 38(4):1086–1095.

Ahlholm, J. U., Helander, M., Lehtimäki, S., Wäli, P., and Saikkonen, K. (2002). Vertically transmitted fungal endophytes: different responses of host-parasite systems to environmental conditions. *Oikos*, 99(1):173–183.

Amijee, F., Tinker, P., and Stribley, D. (1989). The development of endomycorrhizal root systems: Vii. a detailed study of effects of soil phosphorus on colonization. *New Phytologist*, 111(3):435–446.

Atala, C., Acuña-Rodríguez, I. S., Torres-Díaz, C., and Molina-Montenegro, M. A. (2022). Fungal endophytes improve the performance of host plants but do not eliminate the growth/defence trade-off. *New Phytologist*, 235(2).

Bastías, D. A. and Gundel, P. E. (2023). Plant stress responses compromise mutualisms with epichloë endophytes. *Journal of Experimental Botany*, 74(1):19–23.

Bastias, D. A., Martínez-Ghersa, M. A., Ballaré, C. L., and Gundel, P. E. (2017). Epichloë fungal endophytes and plant defenses: not just alkaloids. *Trends in Plant Science*, 22(11):939–948.

Bertness, M. D. and Callaway, R. (1994). Positive interactions in communities. *Trends in ecology & evolution*, 9(5):191–193.

Bronstein, J. L. (1994). Conditional outcomes in mutualistic interactions. *Trends in ecology & evolution*, 9(6):214–217.

Bronstein, J. L. (2001). The exploitation of mutualisms. *Ecology letters*, 4(3):277–287.

Cheplick, G. P. and Faeth, S. H. (2009). *Ecology and evolution of the grass-endophyte symbiosis*. OUP USA.

Clay, K. (1988). Fungal endophytes of grasses: a defensive mutualism between plants and fungi. *Ecology*, 69(1):10–16.

Clay, K. (1990). Fungal endophytes of grasses. *Annual review of Ecology and Systematics*, pages 275–297.

Clay, K., Holah, J., and Rudgers, J. A. (2005). Herbivores cause a rapid increase in hereditary symbiosis and alter plant community composition. *Proceedings of the National Academy of Sciences*, 102(35):12465–12470.

Clay, K. and Schardl, C. (2002). Evolutionary origins and ecological consequences of endophyte symbiosis with grasses. *the american naturalist*, 160(S4):S99–S127.

Decunta, F. A., Pérez, L. I., Malinowski, D. P., Molina-Montenegro, M. A., and Gundel, P. E. (2021). A systematic review on the effects of epichloë fungal endophytes on drought tolerance in cool-season grasses. *Frontiers in Plant Science*, 12:644731.

Dietze, M. C., Sala, A., Carbone, M. S., Czimczik, C. I., Mantooth, J. A., Richardson, A. D., and Vargas, R. (2014). Nonstructural carbon in woody plants. *Annual review of plant biology*, 65(1):667–687.

Eaton, C. J., Cox, M. P., and Scott, B. (2011). What triggers grass endophytes to switch from mutualism to pathogenism? *Plant Science*, 180(2):190–195.

Emery, S. M., Thompson, D., and Rudgers, J. A. (2010). Variation in endophyte symbiosis, herbivory and drought tolerance of ammophila breviligulata populations in the great lakes region. *The American Midland Naturalist*, 163(1):186–196.

Faeth, S. H. (2002). Are endophytic fungi defensive plant mutualists? *Oikos*, 98(1):25–36.

Fowler, J. C., Donald, M. L., Bronstein, J. L., and Miller, T. E. (2023). The geographic footprint of mutualism: How mutualists influence species' range limits. *Ecological Monographs*, 93(1):e1558.

Fowler, J. C., Moutouama, J., and Miller, T. E. (2025). Increasing prevalence of plant-fungal symbiosis across two centuries of environmental change. *Global Change Biology*, 31(11):e70577.

GBIF (2025a). Occurrence download for *Agrostis hyemalis*. Accessed 5 November 2025.

GBIF (2025b). Occurrence download for *Elymus virginicus*. Accessed 5 November 2025.

GBIF (2025c). Occurrence download for *Poa autumnalis*. Accessed 5 November 2025.

Gelman, A. and Hill, J. (2007). *Data analysis using regression and multilevel/hierarchical models*. Cambridge university press.

Giauque, H., Connor, E. W., and Hawkes, C. V. (2019). Endophyte traits relevant to stress tolerance, resource use and habitat of origin predict effects on host plants. *New Phytologist*, 221(4):2239–2249.

Gibert, A., Volaire, F., Barre, P., and Hazard, L. (2012). A fungal endophyte reinforces population adaptive differentiation in its host grass species. *New Phytologist*, 194(2):561–571.

Gundel, P. E., Rudgers, J. A., and Ghersa, C. M. (2011). Incorporating the process of vertical transmission into understanding of host–symbiont dynamics. *Oikos*, 120(8):1121–1128.

Gundel, P. E., Sun, P., Charlton, N. D., Young, C. A., Miller, T. E., and Rudgers, J. A. (2020). Simulated folivory increases vertical transmission of fungal endophytes that deter herbivores and alter tolerance to herbivory in poa autumnalis. *Annals of Botany*, 125(6):981–991.

Gundel, P. E., Ueno, A. C., Casas, C., Miller, T. E., Pérez, L. I., Cuyeu, R., and Omacini, M. (2024). Temporal host-symbiont dynamics in community contexts: Impacts of host fitness and vertical transmission efficiency on symbiosis prevalence. *Functional Ecology*, 38(12):2610–2622.

Hammarlund, S. P. and Harcombe, W. R. (2019). Refining the stress gradient hypothesis in a microbial community. *Proceedings of the National Academy of Sciences*, 116(32):15760–15762.

Hart, E. M. and Bell, K. (2015). *prism: Download data from the Oregon prism project*. R package version 0.0.6.

Hoang, K. L., Salguero-Gómez, R., Pike, V. L., and King, K. C. (2024). The impacts of host association and perturbation on symbiont fitness. *Symbiosis*, 92(3):439–451.

Hoeksema, J. D. and Bruna, E. M. (2015). Context-dependent outcomes of mutualistic interactions. *Mutualism*, 10:181–202.

Holmgren, M. and Scheffer, M. (2010). Strong facilitation in mild environments: the stress gradient hypothesis revisited. *Journal of Ecology*, 98(6):1269–1275.

Jackson, D. and Johnson-Cicalese, J. (1988). A rapid staining method for detection of endophytic fungi in turf and forage grasses. *Phytopathology*, 78:237–239.

Johnson, N. C., Graham, J. H., and Smith, F. A. (1997). Functioning of mycorrhizal associations along the mutualism–parasitism continuum. *The New Phytologist*, 135(4):575–585.

Klironomos, J. N. (2003). Variation in plant response to native and exotic arbuscular mycorrhizal fungi. *Ecology*, 84(9):2292–2301.

Koh, S., Vicari, M., Ball, J., Rakocevic, T., Zaheer, S., Hik, D., and Bazely, D. (2006). Rapid detection of fungal endophytes in grasses for large-scale studies. *Functional Ecology*, pages 736–742.

Leuchtmann, A., Bacon, C. W., Schardl, C. L., White Jr, J. F., and Tadych, M. (2014). Nomenclatural realignment of neotyphodium species with genus epichloë. *Mycologia*, 106(2):202–215.

Liu, M. (2023). Fungi canadenses no. 351: Epichloe elymi. *Canadian Journal of Plant Pathology*, 45(3):300–303.

Lortie, C. J. and Callaway, R. M. (2006). Re-analysis of meta-analysis: support for the stress-gradient hypothesis. *Journal of Ecology*, 94(1):7–16.

Louthan, A. M., Doak, D. F., and Angert, A. L. (2015). Where and when do species interactions set range limits? *Trends in ecology & evolution*, 30(12):780–792.

Oberhofer, M., Güsewell, S., and Leuchtmann, A. (2014). Effects of natural hybrid and non-hybrid epichloë endophytes on the response of hordelymus europaeus to drought stress. *New Phytologist*, 201(1):242–253.

Piccardi, P., Vessman, B., and Mitri, S. (2019). Toxicity drives facilitation between 4 bacterial species. *Proceedings of the National Academy of Sciences*, 116(32):15979–15984.

Porter, S. S., Bantay, R., Friel, C. A., Garoutte, A., Gdanetz, K., Ibarreta, K., Moore, B. M., Shetty, P., Siler, E., and Friesen, M. L. (2020). Beneficial microbes ameliorate abiotic and biotic sources of stress on plants. *Functional Ecology*, 34(10):2075–2086.

Pringle, E. G. (2016). Integrating plant carbon dynamics with mutualism ecology. *New Phytologist*, 210(1):71–75.

R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Reynolds, H. L., Packer, A., Bever, J. D., and Clay, K. (2003). Grassroots ecology: plant–microbe–soil interactions as drivers of plant community structure and dynamics. *Ecology*, 84(9):2281–2291.

Rodriguez, R., White Jr, J., Arnold, A. E., and Redman, a. R. a. (2009). Fungal endophytes: diversity and functional roles. *New phytologist*, 182(2):314–330.

Rudgers, J. A., Afkhami, M. E., Bell-Dereske, L., Chung, Y. A., Crawford, K. M., Kivlin, S. N., Mann, M. A., and Nuñez, M. A. (2020). Climate disruption of plant-microbe interactions. *Annual review of ecology, evolution, and systematics*, 51:561–586.

Rudgers, J. A., Fletcher, R. A., Olivas, E., Young, C. A., Charlton, N. D., Pearson, D. E., and Maron, J. L. (2016). Long-term ungulate exclusion reduces fungal symbiont prevalence in native grasslands. *Oecologia*, 181(4):1151–1161.

Rudgers, J. A., Miller, T. E., Ziegler, S. M., and Craven, K. D. (2012). There are many ways to be a mutualist: endophytic fungus reduces plant survival but increases population growth. *Ecology*, 93(3):565–574.

Ruiz-lazona, J. M., Azcón, R., and Palma, J. (1996). Superoxide dismutase activity in arbuscular mycorrhizal lactuca sativa plants subjected to drought stress. *New Phytologist*, 134(2):327–333.

Saikkonen, K., Saari, S., and Helander, M. (2010). Defensive mutualism between plants and endophytic fungi? *Fungal Diversity*, 41(1):101–113.

Saikkonen, K., Wäli, P., Helander, M., and Faeth, S. H. (2004). Evolution of endophyte–plant symbioses. *Trends in plant science*, 9(6):275–280.

Schardl, C. L., Leuchtmann, A., and Spiering, M. J. (2004). Symbioses of grasses with seedborne fungal endophytes. *Annu. Rev. Plant Biol.*, 55(1):315–340.

Shaw, R. B. (2011). *Guide to Texas grasses*. Texas A&M University Press.

Soudzilovskaia, N. A., van der Heijden, M. G., Cornelissen, J. H., Makarov, M. I., Onipchenko, V. G., Maslov, M. N., Akhmetzhanova, A. A., and van Bodegom, P. M. (2015). Quantitative assessment of the differential impacts of arbuscular and ectomycorrhiza on soil carbon cycling. *New Phytologist*, 208(1):280–293.

Stan Development Team (2024). RStan: the R interface to Stan. R package version 2.32.6.

Thirkell, T. J., Pastok, D., and Field, K. J. (2020). Carbon for nutrient exchange between arbuscular mycorrhizal fungi and wheat varies according to cultivar and changes in atmospheric carbon dioxide concentration. *Global change biology*, 26(3):1725–1738.

Toby Kiers, E., Palmer, T. M., Ives, A. R., Bruno, J. F., and Bronstein, J. L. (2010). Mutualisms in a changing world: an evolutionary perspective. *Ecology letters*, 13(12):1459–1474.

Vega, F. E. (2008). Insect pathology and fungal endophytes. *Journal of invertebrate pathology*, 98(3):277–279.

Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and computing*, 27:1413–1432.

White Jr, J. F. (1994). Endophyte-host associations in grasses. xx. structural and reproductive studies of epichloë amarillans sp. nov. and comparisons to e. typhina. *Mycologia*, 86(4):571–580.

White Jr, J. F. and Torres, M. S. (2010). Is plant endophyte-mediated defensive mutualism the result of oxidative stress protection? *Physiologia Plantarum*, 138(4):440–446.

Wilson, A. and Thompson, K. (1989). A comparative study of reproductive allocation in 40 british grasses. *Functional Ecology*, pages 297–302.

Yule, K. M., Miller, T. E., and Rudgers, J. A. (2013). Costs, benefits, and loss of vertically transmitted symbionts affect host population dynamics. *Oikos*, 122(10):1512–1520.

564 # Figures

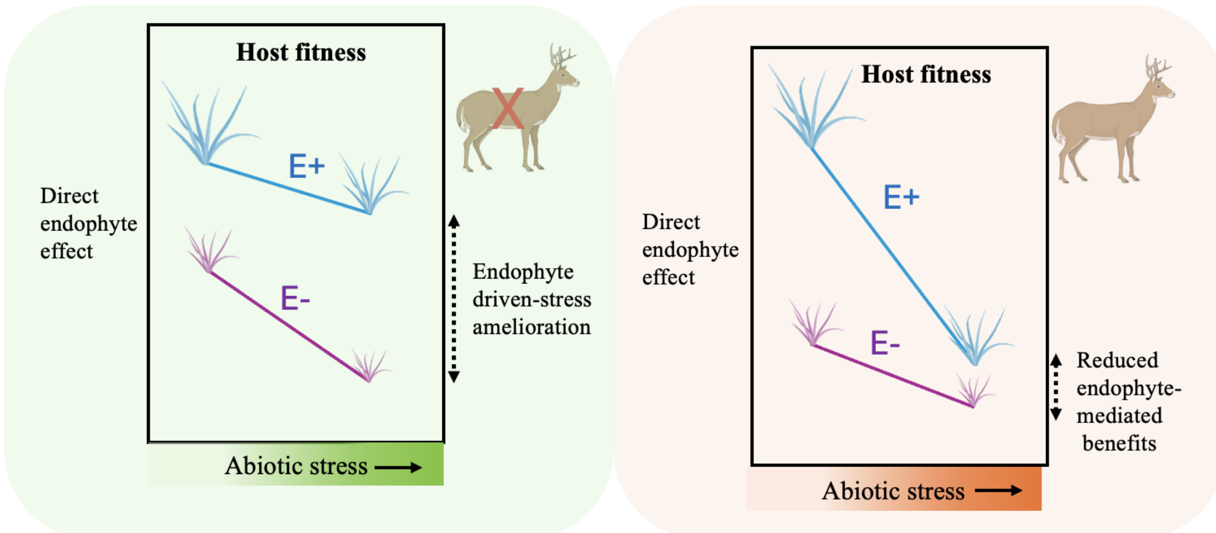

Figure 1: Conceptual framework illustrating the predicted effects of abiotic stress, biotic stress, and endophyte-mediated mutualism on host plant fitness (adapted from (Porter et al., 2020)). (a) **Endophyte-driven stress amelioration scenario:** Endophytes are predicted to enhance host performance under increasing abiotic stress (e.g., low precipitation) in the absence of herbivores by mitigating drought-related physiological costs. (b) **Reduced endophyte-mediated benefits scenario:** When abiotic stress is combined with herbivory, the positive effect of endophytes is expected to weaken. Limited plant resources are divided between defense against herbivores and sustaining the endophyte, reducing the endophyte's capacity to ameliorate drought stress and potentially resulting in a net cost of hosting an endophyte. Colors are used solely for visualization: blue lines represent endophyte-infected plants and violet lines represent endophyte-free plants.

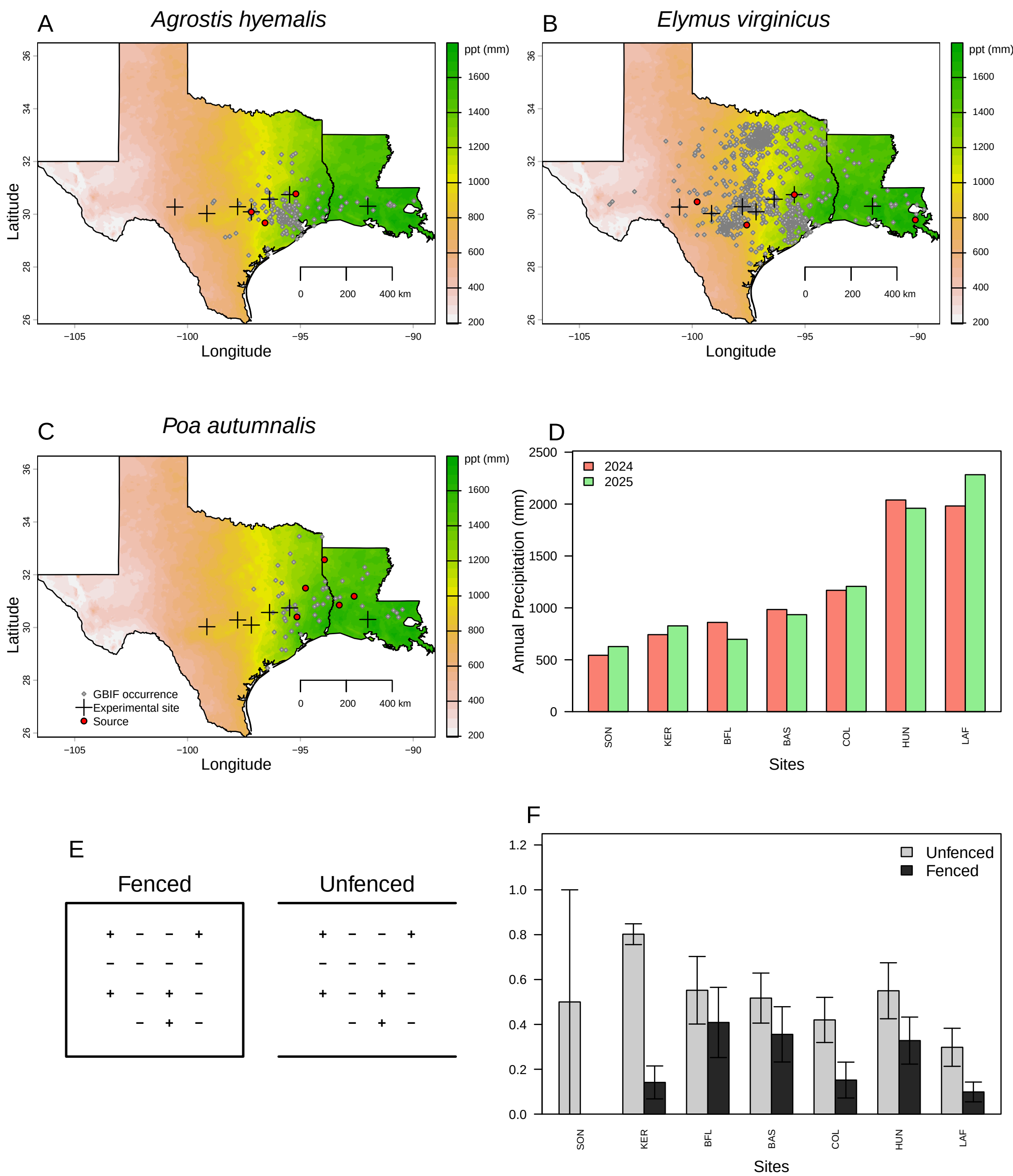

Figure 2: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1996–2025) from PRISM data (Hart and Bell, 2015), illustrating the precipitation gradient across experimental sites (scale on the right is in mm). Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). (D) Annual cumulative precipitation during each census year at each experimental site, ordered west to east. Values are also derived from PRISM climate data. Note that Kerville (KER) was not censused in 2025. (E) Schematic illustration of fenced (left) and unfenced (right) plots used to assess herbivory. Symbols (+/−) represent plant endophyte status, with "+" indicating endophyte-positive and "−" indicating endophyte-free individuals; plot shape corresponds to experimental design. (F) Proportion of damaged plants in fenced (salmon) and unfenced (light green) plots across experimental sites, with mean ± standard error. Sites are ordered west to east as in panel D. Panels E–F illustrate herbivory treatments and outcomes, and species-specific breakdowns are provided in Supplementary Figures.

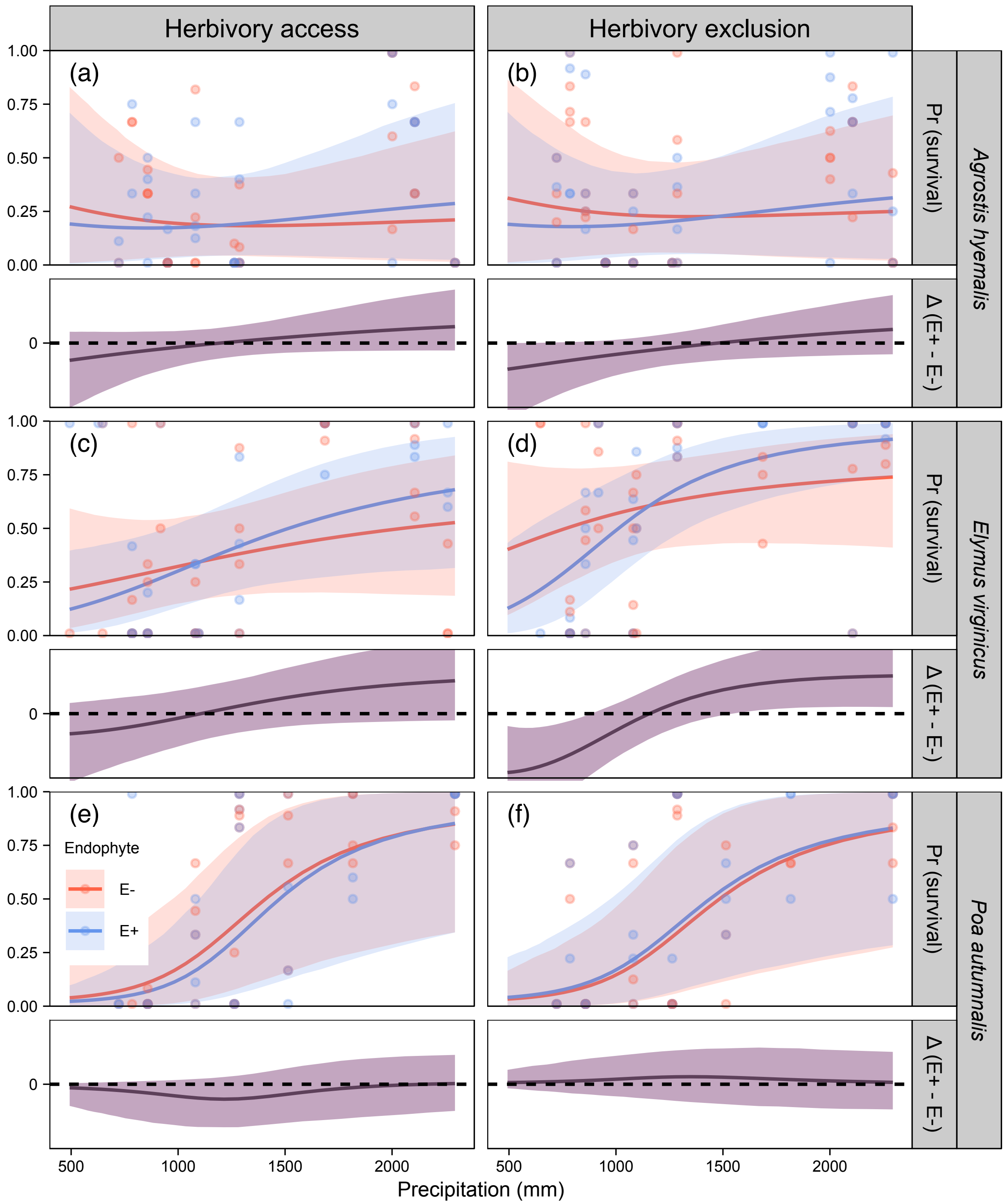

Figure 3: Predicted survival rate (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

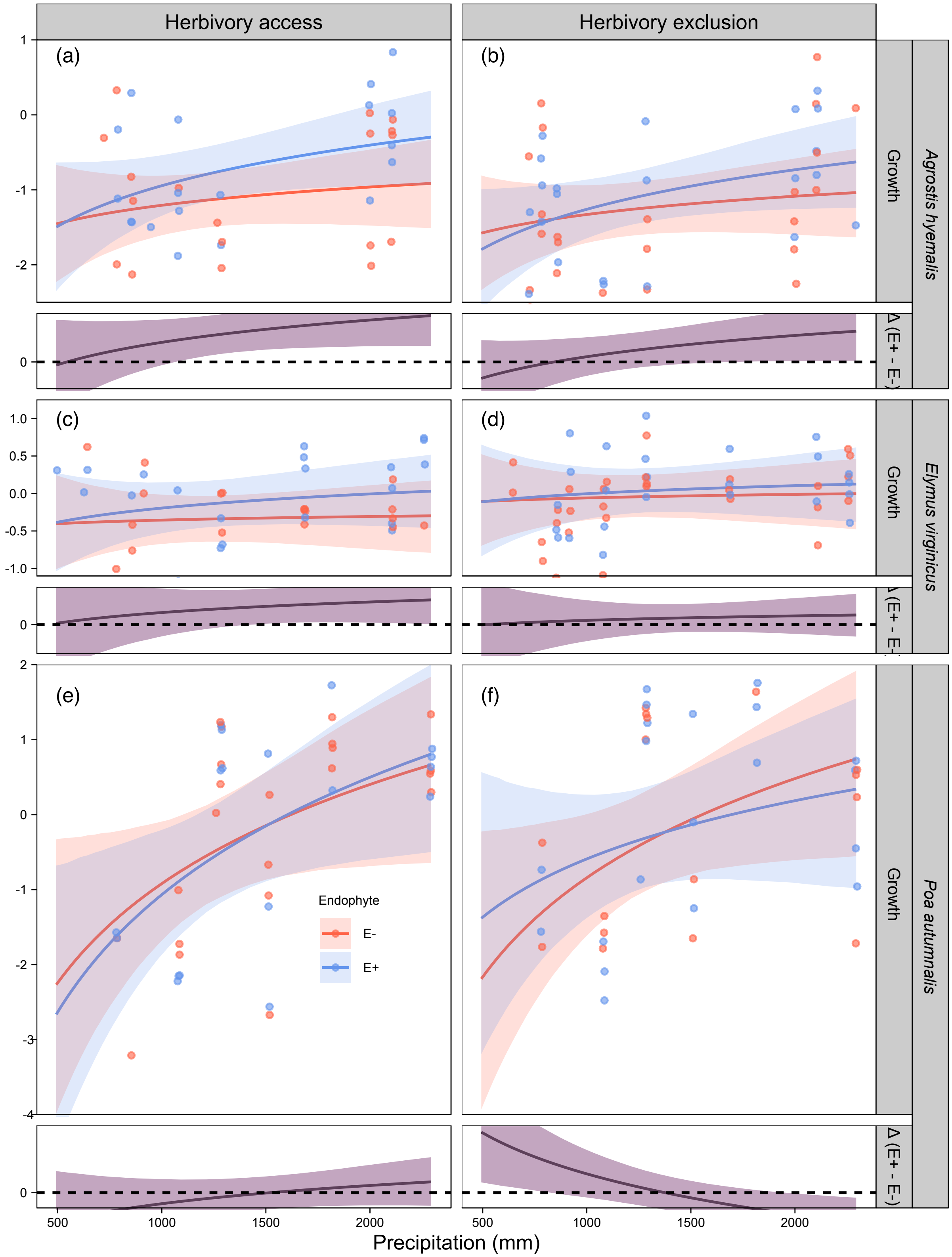

Figure 4: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

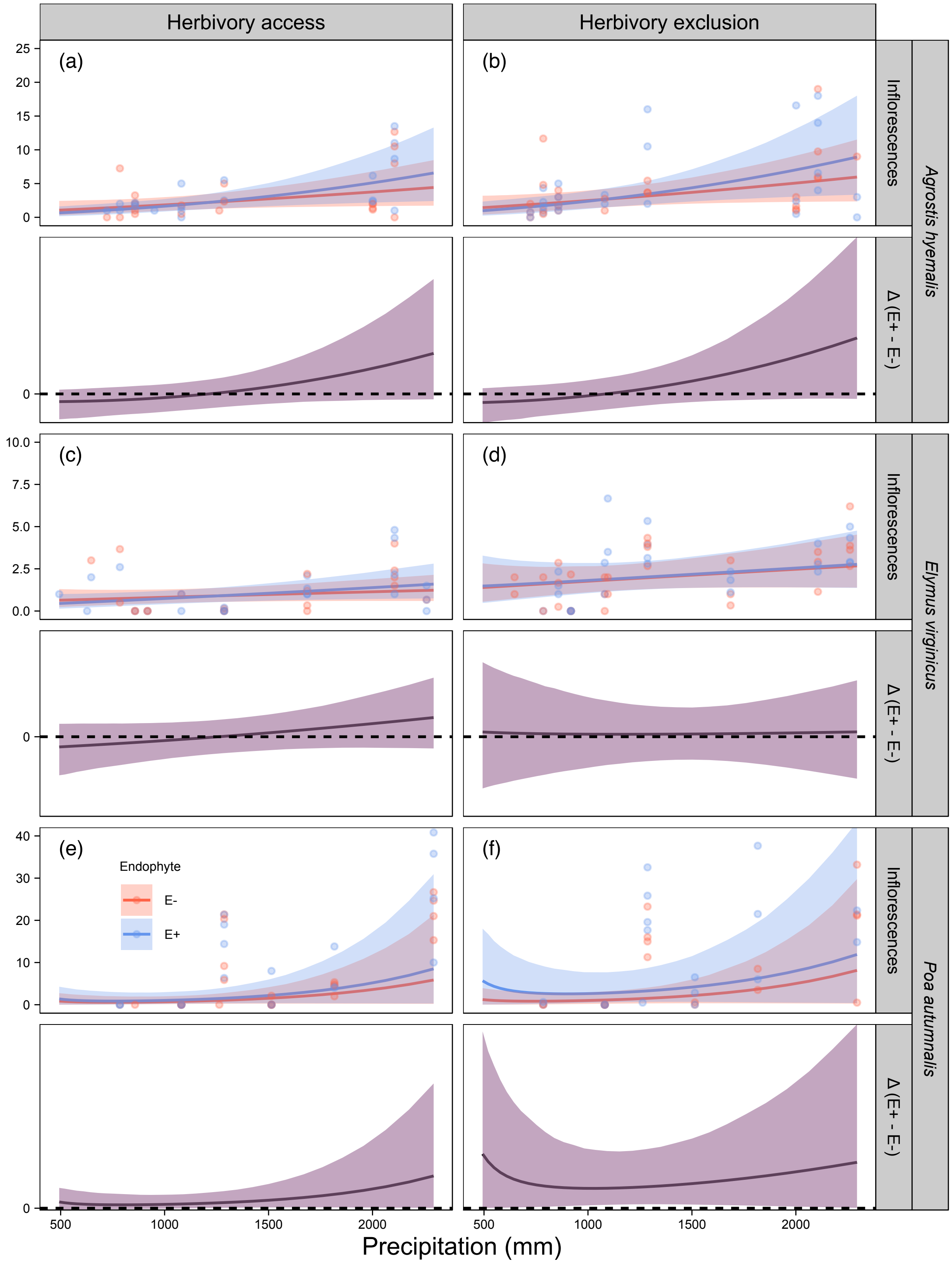

Figure 5: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+–E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means, jittered for visualization. Panels are grouped by species and herbivory treatment (Fenced, Unfenced).

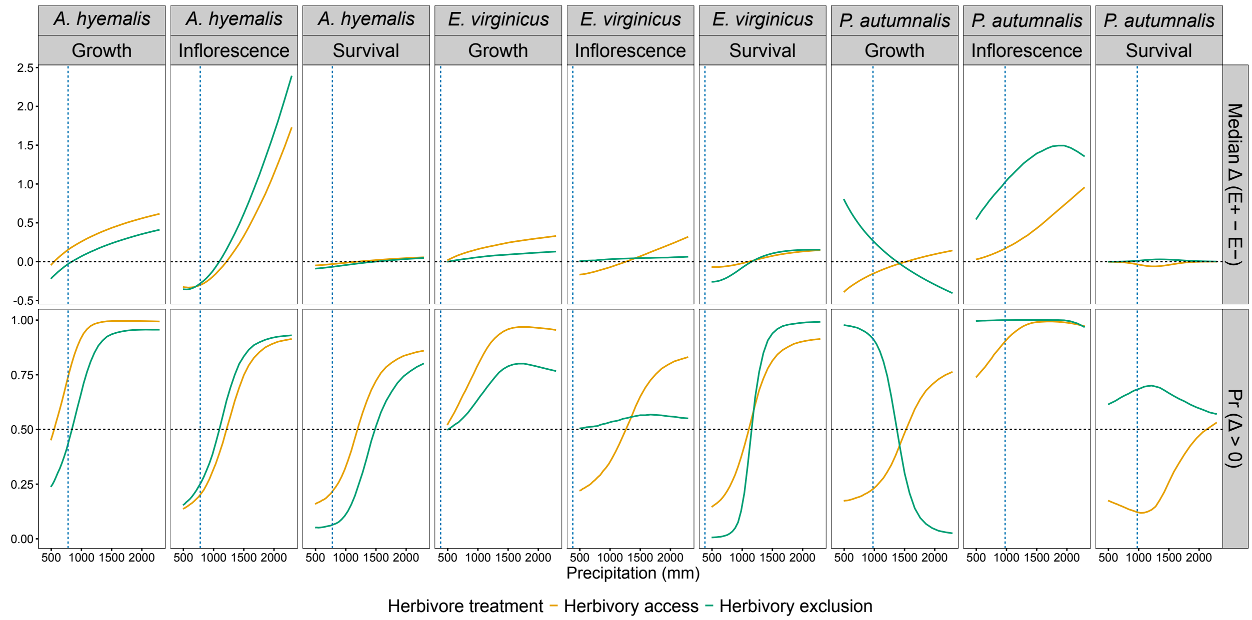

Figure 6: Predicted effects of herbivore exclusion on survival, growth, and inflorescence production across precipitation gradients for *A. hyemalis*, *E. virginicus*, and *P. autumnalis*. Panels show the median difference between fenced and unfenced treatments ($\Delta = E^{+} - E^{-}$) and the probability that this difference is positive ($\Pr[\Delta > 0]$). Horizontal dashed lines indicate $\Delta = 0$ and $\Pr(\Delta > 0) = 0.5$, and vertical dashed lines denote species-specific mean annual precipitation at range limits. Colors indicate herbivore exclusion treatments.

Tables

Table 1: Comparison of models assessing the drivers of grass–endophyte symbiosis across four fitness components (vital rates). ΔELPD represents the difference in expected log predictive density relative to the best-fitting model, and SE denotes the standard error of this difference.

Vital rate	**Model type**	**ΔELPD**	**SE**
Survival	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.33	2.60
	Endophyte × Climate × Herbivory	−1.92	1.43
	Endophyte + Climate + Herbivory	−4.98	3.24
Growth	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−1.72	1.99
	Endophyte × Climate × Herbivory	−2.04	3.05
	Endophyte + Climate + Herbivory	−2.47	3.72
Inflorescence	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−1.65	1.81
	Endophyte × Climate × Herbivory	−3.02	2.65
	Endophyte + Climate + Herbivory	−4.65	2.70
Spikelet	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.56	2.25
	Endophyte × Climate × Herbivory	−0.95	1.54
	Endophyte + Climate + Herbivory	−4.23	1.87

S.1 Supporting Information

S.1.1 Supporting Methods

Candidate models for abiotic and biotic drivers of cool-season grass species

1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and their interaction:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i \\ &+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and their interaction:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i + \beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i] endo_i herb_i \\ &+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the three-way interaction:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i \\ &+ \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i + \beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i] herb_i P_i + \beta_{\mathrm{herb}\times\mathrm{endo}}[\mathrm{Sp}_i] herb_i endo_i \\ &+ \beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i] endo_i herb_i P_i + \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

4. **Endophyte + Climate + Herbivory (main effects only):** includes all three predictors but no interactions:

$$\begin{aligned}\mu_i = &\beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i \\ &+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}\end{aligned}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\mathrm{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\mathrm{endo}}[\mathrm{Sp}_i]$ and $\beta_{\mathrm{herb}}[\mathrm{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively. The linear effect of precipitation is captured by $\beta_P[\mathrm{Sp}_i]$, while interactions are represented by $\beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i]$, and the three-way interaction $\beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i]$. All models include hierarchical random effects for site-year, source population, and plot, with species-specific random effects $\phi_{\mathrm{site}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{site}})$, $\omega_{\mathrm{source}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{source}})$, and $\rho_{\mathrm{plot}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{plot}})$.

S.1.2 Supporting Tables

Table S1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	San Jacinto County, TX
AGHY	COHR	29.672833	-96.567517	Columbus, TX
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve[32]
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS[33]	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University Field Station, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S2: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
Private property in Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S3: Realized numbers of endophyte-free (E−) and endophyte-infected (E+) individuals per species and population, with total number of clones.

Species	**Population**	**E−**	**E+**	**Total clones**
Agrostis hyemalis	COHR	288	290	578
Agrostis hyemalis	LPEF	270	286	556
Agrostis hyemalis	SJC	284	262	546
Elymus virginicus	HUNT	100	22	122
Elymus virginicus	JLP	188	164	352
Elymus virginicus	PALM	384	500	884
Elymus virginicus	RL5	166	96	262
Elymus virginicus	SHS	2	58	60
Poa autumnalis	LA1	142	116	258
Poa autumnalis	LA2	234	272	506
Poa autumnalis	LA7	102	80	182
Poa autumnalis	SFAEF	242	252	494

Table S4: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0.00	0.00
Survival	Linear	-0.50	0.84
Growth	Quadratic	0.00	0.00
Growth	Linear	-0.04	1.03
Inflorescence	Quadratic	0.00	0.00
Inflorescence	Linear	-0.11	0.64
Spikelet	Quadratic	0.00	0.00
Spikelet	Linear	-0.56	0.52

Table S5: Posterior summary of survival ratios (E^+ / E^-) under low-climate stress. Median, 90% credible intervals, and posterior probability that E^+ performs worse than E^- are shown for each species and herbivory treatment.

Species	Herbivory	Median ratio	Lower 90%	Upper 90%	Prob(E^+ worse)
Agrostis hyemalis	Herbivory access	0.666	0.253	1.36	0.825
Agrostis hyemalis	Herbivory exclusion	0.544	0.221	1.01	0.946
Elymus virginicus	Herbivory access	0.602	0.202	1.44	0.831
Elymus virginicus	Herbivory exclusion	0.352	0.0975	0.801	0.991
Poa autumnalis	Herbivory access	0.405	0.0785	1.83	0.837
Poa autumnalis	Herbivory exclusion	1.28	0.356	4.91	0.371

Table S6: Posterior summaries of Δ ($E^{+} - E^{-}$) for survival across precipitation quartiles for each species and herbivore treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore treatment	Precipitation bin (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Herbivory access	493	-0.049	-0.302	0.053	0.158
Agrostis hyemalis	Herbivory access	678	-0.038	-0.215	0.052	0.187
Agrostis hyemalis	Herbivory access	1036	-0.012	-0.099	0.058	0.357
Agrostis hyemalis	Herbivory access	1424	0.015	-0.050	0.102	0.681
Agrostis hyemalis	Herbivory access	2294	0.056	-0.035	0.249	0.860
Agrostis hyemalis	Herbivory exclusion	493	-0.090	-0.358	0.000	0.051
Agrostis hyemalis	Herbivory exclusion	678	-0.076	-0.264	0.002	0.057
Agrostis hyemalis	Herbivory exclusion	1036	-0.041	-0.142	0.019	0.127
Agrostis hyemalis	Herbivory exclusion	1424	-0.004	-0.085	0.072	0.448
Agrostis hyemalis	Herbivory exclusion	2294	0.047	-0.052	0.224	0.802
Elymus virginicus	Herbivory access	493	-0.069	-0.321	0.050	0.145
Elymus virginicus	Herbivory access	678	-0.065	-0.254	0.068	0.189
Elymus virginicus	Herbivory access	1036	-0.014	-0.143	0.113	0.424
Elymus virginicus	Herbivory access	1424	0.061	-0.069	0.194	0.779
Elymus virginicus	Herbivory access	2294	0.148	-0.031	0.356	0.914
Elymus virginicus	Herbivory exclusion	493	-0.259	-0.550	-0.058	0.006
Elymus virginicus	Herbivory exclusion	678	-0.230	-0.430	-0.058	0.012
Elymus virginicus	Herbivory exclusion	1036	-0.059	-0.183	0.061	0.210
Elymus virginicus	Herbivory exclusion	1424	0.090	-0.022	0.225	0.907
Elymus virginicus	Herbivory exclusion	2294	0.154	0.031	0.400	0.992
Poa autumnalis	Herbivory access	493	-0.001	-0.101	0.005	0.175
Poa autumnalis	Herbivory access	678	-0.005	-0.143	0.010	0.154
Poa autumnalis	Herbivory access	1036	-0.043	-0.196	0.022	0.118
Poa autumnalis	Herbivory access	1424	-0.051	-0.188	0.060	0.214
Poa autumnalis	Herbivory access	2294	0.000	-0.124	0.137	0.532
Poa autumnalis	Herbivory exclusion	493	0.000	-0.019	0.067	0.614
Poa autumnalis	Herbivory exclusion	678	0.001	-0.033	0.092	0.641
Poa autumnalis	Herbivory exclusion	1036	0.016	-0.058	0.136	0.689
Poa autumnalis	Herbivory exclusion	1424	0.029	-0.088	0.166	0.679
Poa autumnalis	Herbivory exclusion	2294	0.002	-0.117	0.151	0.570

Table S7: Posterior summaries of Δ ($E^{+} - E^{-}$) for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	Herbivore treatment	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Herbivory access	493	-0.046	-0.643	0.561	0.450
Agrostis hyemalis	Herbivory access	678	0.092	-0.352	0.543	0.629
Agrostis hyemalis	Herbivory access	1036	0.274	-0.019	0.571	0.939
Agrostis hyemalis	Herbivory access	1424	0.412	0.147	0.683	0.994
Agrostis hyemalis	Herbivory access	2294	0.616	0.212	1.022	0.994
Agrostis hyemalis	Herbivory exclusion	493	-0.221	-0.723	0.290	0.237
Agrostis hyemalis	Herbivory exclusion	678	-0.090	-0.466	0.284	0.347
Agrostis hyemalis	Herbivory exclusion	1036	0.084	-0.165	0.336	0.710
Agrostis hyemalis	Herbivory exclusion	1424	0.214	-0.034	0.460	0.925
Agrostis hyemalis	Herbivory exclusion	2294	0.411	0.017	0.799	0.956
Elymus virginicus	Herbivory access	493	0.017	-0.618	0.647	0.520
Elymus virginicus	Herbivory access	678	0.084	-0.400	0.566	0.614
Elymus virginicus	Herbivory access	1036	0.168	-0.137	0.476	0.814
Elymus virginicus	Herbivory access	1424	0.232	-0.002	0.470	0.948
Elymus virginicus	Herbivory access	2294	0.330	0.008	0.643	0.955
Elymus virginicus	Herbivory exclusion	493	-0.002	-0.594	0.578	0.498
Elymus virginicus	Herbivory exclusion	678	0.023	-0.419	0.457	0.532
Elymus virginicus	Herbivory exclusion	1036	0.060	-0.205	0.316	0.651
Elymus virginicus	Herbivory exclusion	1424	0.087	-0.101	0.274	0.777
Elymus virginicus	Herbivory exclusion	2294	0.130	-0.160	0.410	0.767
Poa autumnalis	Herbivory access	493	-0.393	-1.089	0.289	0.174
Poa autumnalis	Herbivory access	678	-0.283	-0.803	0.230	0.185
Poa autumnalis	Herbivory access	1036	-0.136	-0.448	0.179	0.242
Poa autumnalis	Herbivory access	1424	-0.024	-0.245	0.202	0.431
Poa autumnalis	Herbivory access	2294	0.143	-0.175	0.474	0.763
Poa autumnalis	Herbivory exclusion	493	0.806	0.137	1.478	0.977
Poa autumnalis	Herbivory exclusion	678	0.556	0.058	1.060	0.967
Poa autumnalis	Herbivory exclusion	1036	0.223	-0.077	0.529	0.890
Poa autumnalis	Herbivory exclusion	1424	-0.028	-0.251	0.199	0.418
Poa autumnalis	Herbivory exclusion	2294	-0.406	-0.737	-0.067	0.025

Table S8: Posterior summaries of Δ ($E^+ - E^-$) for flowering output at precipitation quantiles for each species and herbivore treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore treatment	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
Species	Herbivory	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
Agrostis hyemalis	Herbivory access	493	-0.325	-1.371	0.229	0.137
Agrostis hyemalis	Herbivory access	678	-0.327	-1.230	0.320	0.172
Agrostis hyemalis	Herbivory access	1036	-0.157	-0.929	0.557	0.344
Agrostis hyemalis	Herbivory access	1424	0.231	-0.589	1.219	0.691
Agrostis hyemalis	Herbivory access	2294	1.730	-0.289	6.251	0.914
Agrostis hyemalis	Herbivory exclusion	493	-0.355	-1.550	0.311	0.154
Agrostis hyemalis	Herbivory exclusion	678	-0.332	-1.339	0.426	0.206
Agrostis hyemalis	Herbivory exclusion	1036	-0.067	-0.897	0.796	0.439
Agrostis hyemalis	Herbivory exclusion	1424	0.481	-0.492	1.772	0.800
Agrostis hyemalis	Herbivory exclusion	2294	2.394	-0.264	8.520	0.930
Elymus virginicus	Herbivory access	493	-0.167	-0.742	0.249	0.220
Elymus virginicus	Herbivory access	678	-0.142	-0.606	0.255	0.250
Elymus virginicus	Herbivory access	1036	-0.062	-0.410	0.274	0.369
Elymus virginicus	Herbivory access	1424	0.046	-0.277	0.391	0.606
Elymus virginicus	Herbivory access	2294	0.320	-0.226	1.141	0.831
Elymus virginicus	Herbivory exclusion	493	0.006	-1.001	1.434	0.505
Elymus virginicus	Herbivory exclusion	678	0.015	-0.824	1.125	0.512
Elymus virginicus	Herbivory exclusion	1036	0.033	-0.576	0.732	0.536
Elymus virginicus	Herbivory exclusion	1424	0.044	-0.445	0.566	0.561
Elymus virginicus	Herbivory exclusion	2294	0.064	-0.808	1.089	0.551
Poa autumnalis	Herbivory access	493	0.029	-0.152	1.709	0.738
Poa autumnalis	Herbivory access	678	0.070	-0.097	1.287	0.798
Poa autumnalis	Herbivory access	1036	0.194	-0.029	1.143	0.921
Poa autumnalis	Herbivory access	1424	0.385	0.054	1.767	0.987
Poa autumnalis	Herbivory access	2294	0.957	0.027	10.518	0.972
Poa autumnalis	Herbivory exclusion	493	0.543	0.021	14.821	0.996
Poa autumnalis	Herbivory exclusion	678	0.744	0.072	8.201	0.998
Poa autumnalis	Herbivory exclusion	1036	1.076	0.259	4.875	1.000
Poa autumnalis	Herbivory exclusion	1424	1.349	0.322	5.461	1.000
Poa autumnalis	Herbivory exclusion	2294	1.355	0.031	15.437	0.968

Table S9: Posterior summaries of Δ ($E^{+} - E^{-}$) at precipitation quantiles for each species and herbivore treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore treatment	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Elymus virginicus	Herbivory access	493	-3.149	-11.141	3.194	0.202
Elymus virginicus	Herbivory access	678	-2.375	-8.360	2.851	0.224
Elymus virginicus	Herbivory access	1036	-1.132	-5.122	2.668	0.303
Elymus virginicus	Herbivory access	1424	-0.095	-3.238	3.106	0.481
Elymus virginicus	Herbivory access	2294	1.676	-2.131	6.291	0.770
Elymus virginicus	Herbivory exclusion	493	-4.550	-11.872	1.236	0.092
Elymus virginicus	Herbivory exclusion	678	-3.364	-8.645	1.306	0.111
Elymus virginicus	Herbivory exclusion	1036	-1.444	-4.572	1.644	0.211
Elymus virginicus	Herbivory exclusion	1424	0.261	-2.038	2.652	0.577
Elymus virginicus	Herbivory exclusion	2294	3.229	-0.389	8.420	0.929
Poa autumnalis	Herbivory access	493	-0.584	-6.085	3.143	0.320
Poa autumnalis	Herbivory access	678	-0.584	-4.893	2.756	0.330
Poa autumnalis	Herbivory access	1036	-0.408	-3.240	2.226	0.373
Poa autumnalis	Herbivory access	1424	-0.025	-2.094	2.056	0.491
Poa autumnalis	Herbivory access	2294	1.097	-2.362	5.455	0.719
Poa autumnalis	Herbivory exclusion	493	2.630	-1.310	12.708	0.890
Poa autumnalis	Herbivory exclusion	678	2.943	-0.645	10.507	0.917
Poa autumnalis	Herbivory exclusion	1036	3.267	0.227	7.905	0.963
Poa autumnalis	Herbivory exclusion	1424	3.333	0.797	6.547	0.987
Poa autumnalis	Herbivory exclusion	2294	2.748	-1.344	8.544	0.871

Table S10: Posterior summary of spikelet ratios (E^+ / E^-) under low-climate stress. Median, 90% credible intervals, and posterior probability that E^+ outperforms E^- are shown for each species and herbivory treatment.

Species	Herbivory	Median ratio	Lower 90%	Upper 90%	Prob(E^+ worse)
Agrostis hyemalis	Herbivory access	0.835	0.566	1.21	0.785
Agrostis hyemalis	Herbivory exclusion	0.790	0.569	1.07	0.897
Elymus virginicus	Herbivory access	0.900	0.612	1.32	0.675
Elymus virginicus	Herbivory exclusion	1.35	0.928	1.98	0.0942

595 ### S.1.3 Supporting Figures

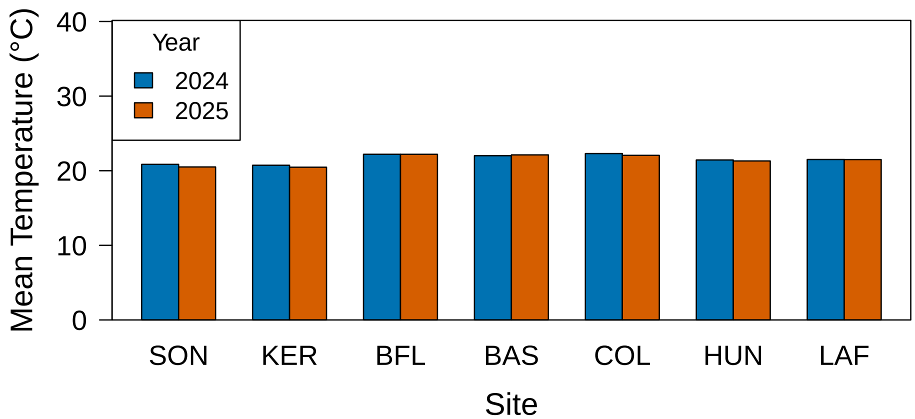

Fig S1: Annual temperature (°C) for 2024 and 2025 census. Sites are arranged from west to east based on longitude.

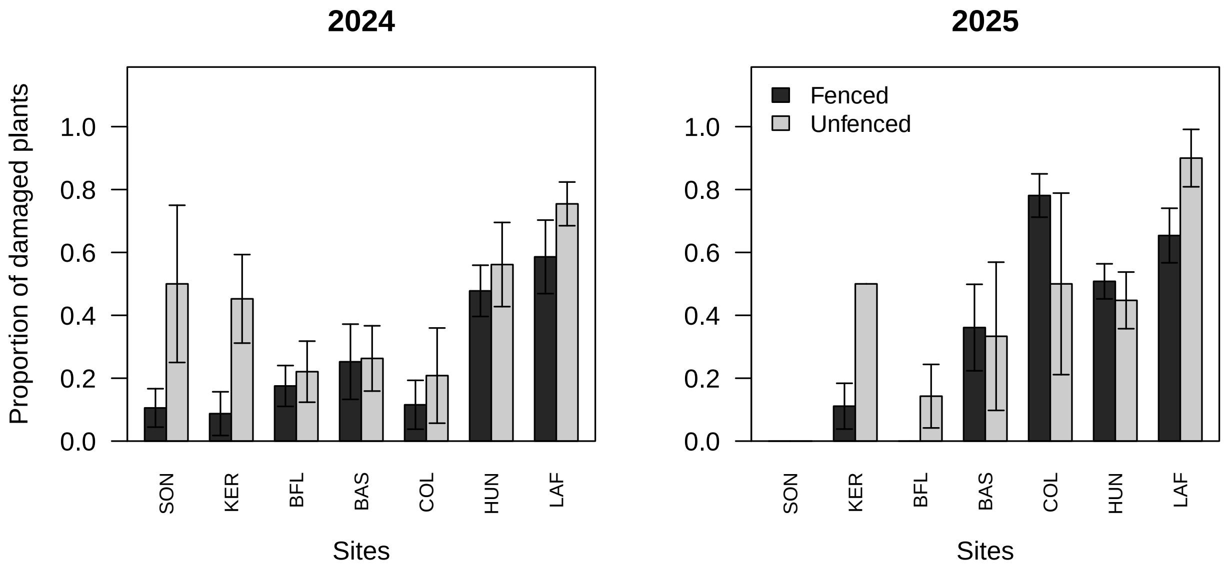

Fig S2: Proportion of plants damaged by herbivory across sites in 2024 and 2025. Barplots show fenced (dark) versus unfenced (grey) plots for each site (ordered west to east). Error bars indicate standard errors across plots within each site. Panels show results for 2024 (left) and 2025 (right).

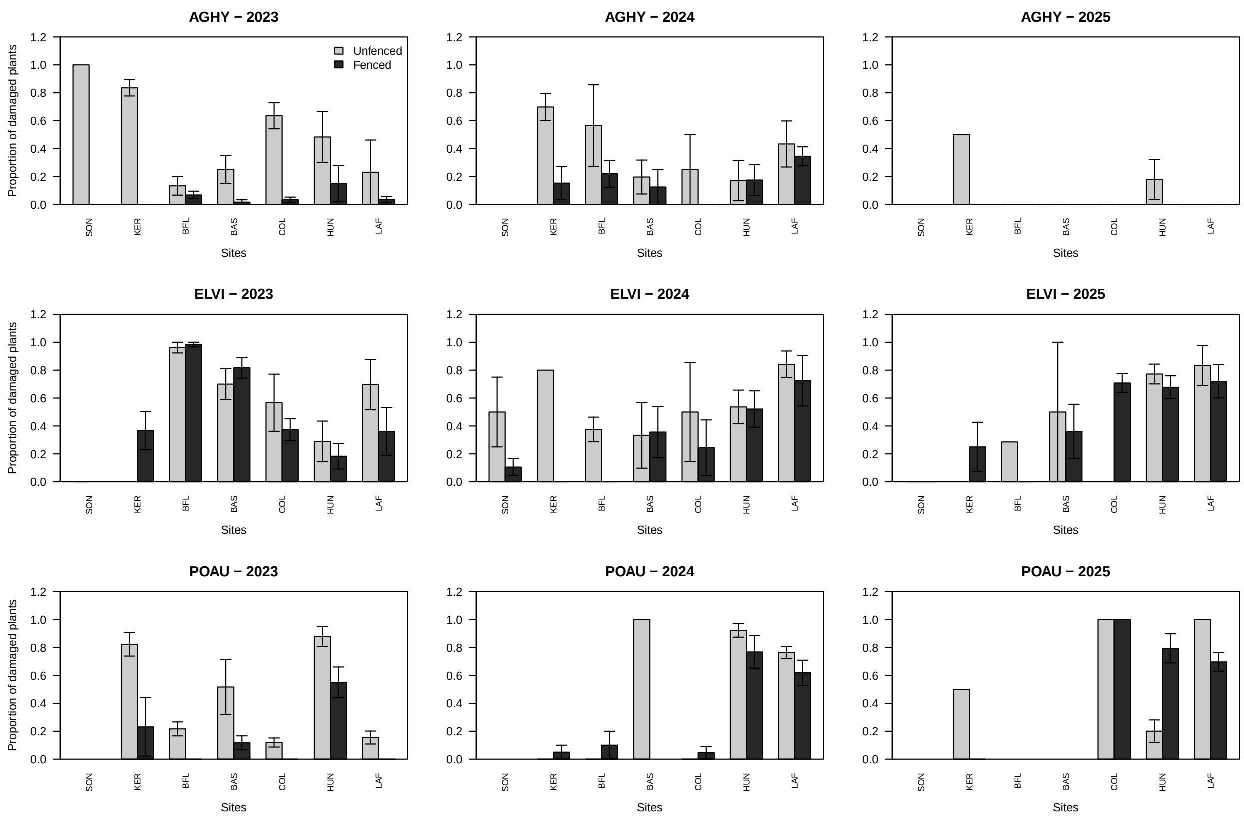

Fig S3: Proportion of plants damaged by herbivory across sites for three species (*Agrostis hyemalis*, *Elymus virginicus*, *Poa autumnalis*) over the years 2023–2025. Barplots show fenced (grey80) versus unfenced (grey15) proportion of damage for each site (ordered west to east: SON, KER, BFL, BAS, COL, HUN, LAF). Error bars indicate standard errors across plots within each site. Panels are arranged by species (rows) and year (columns), with the y-axis labeled only for the first column for clarity.

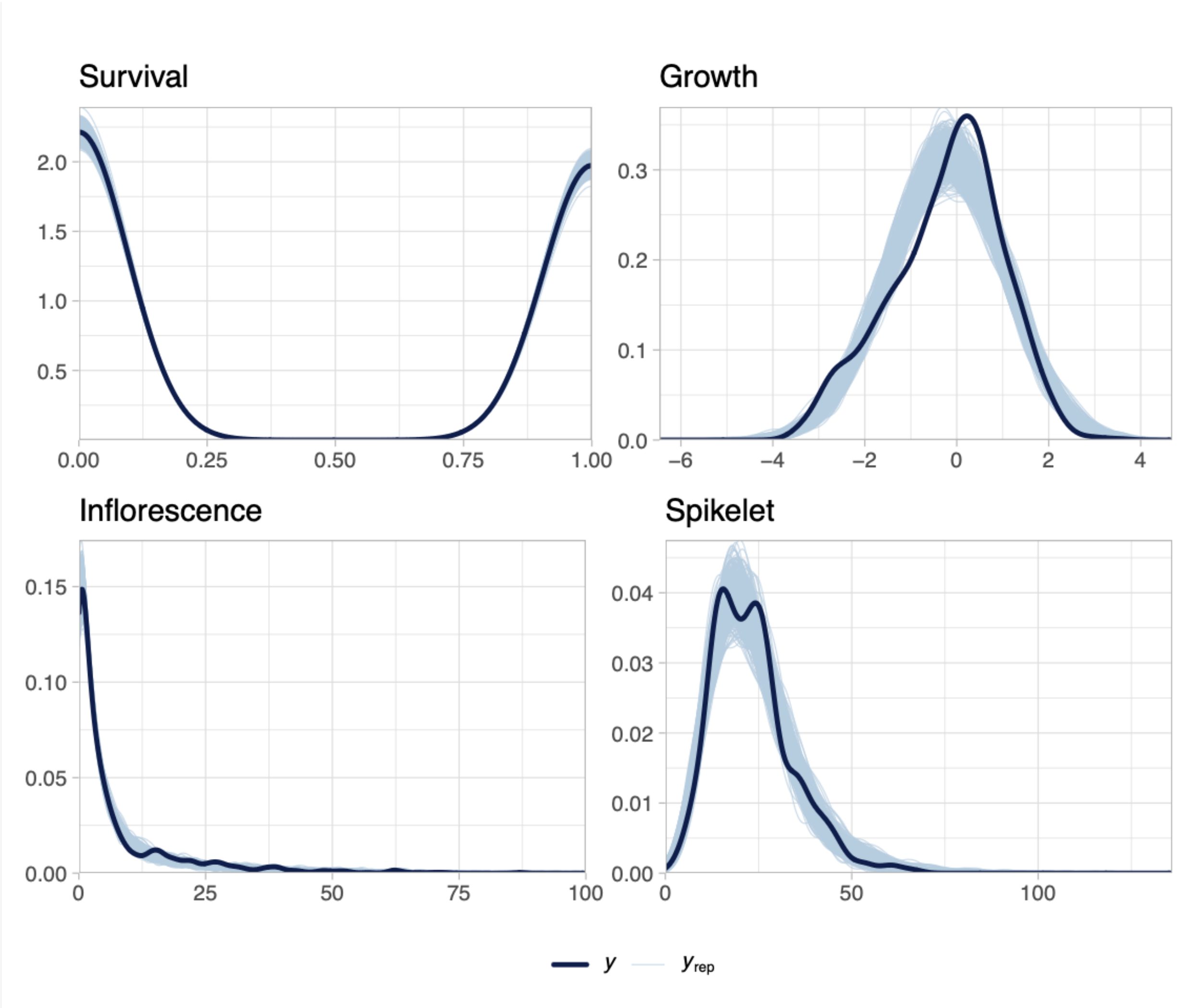

Fig S4: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

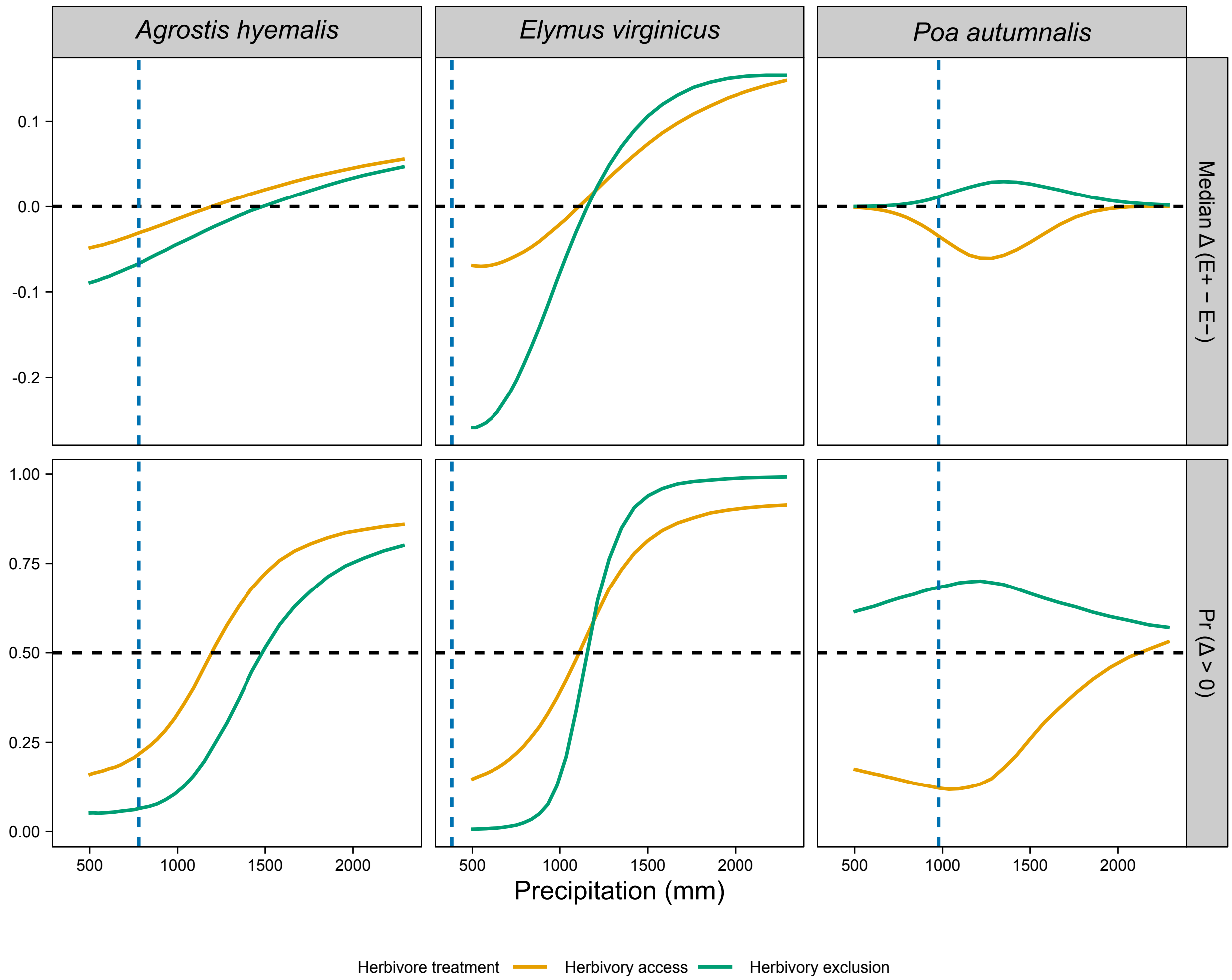

Fig S5: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median Δ ($E^+ - E^-$),or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

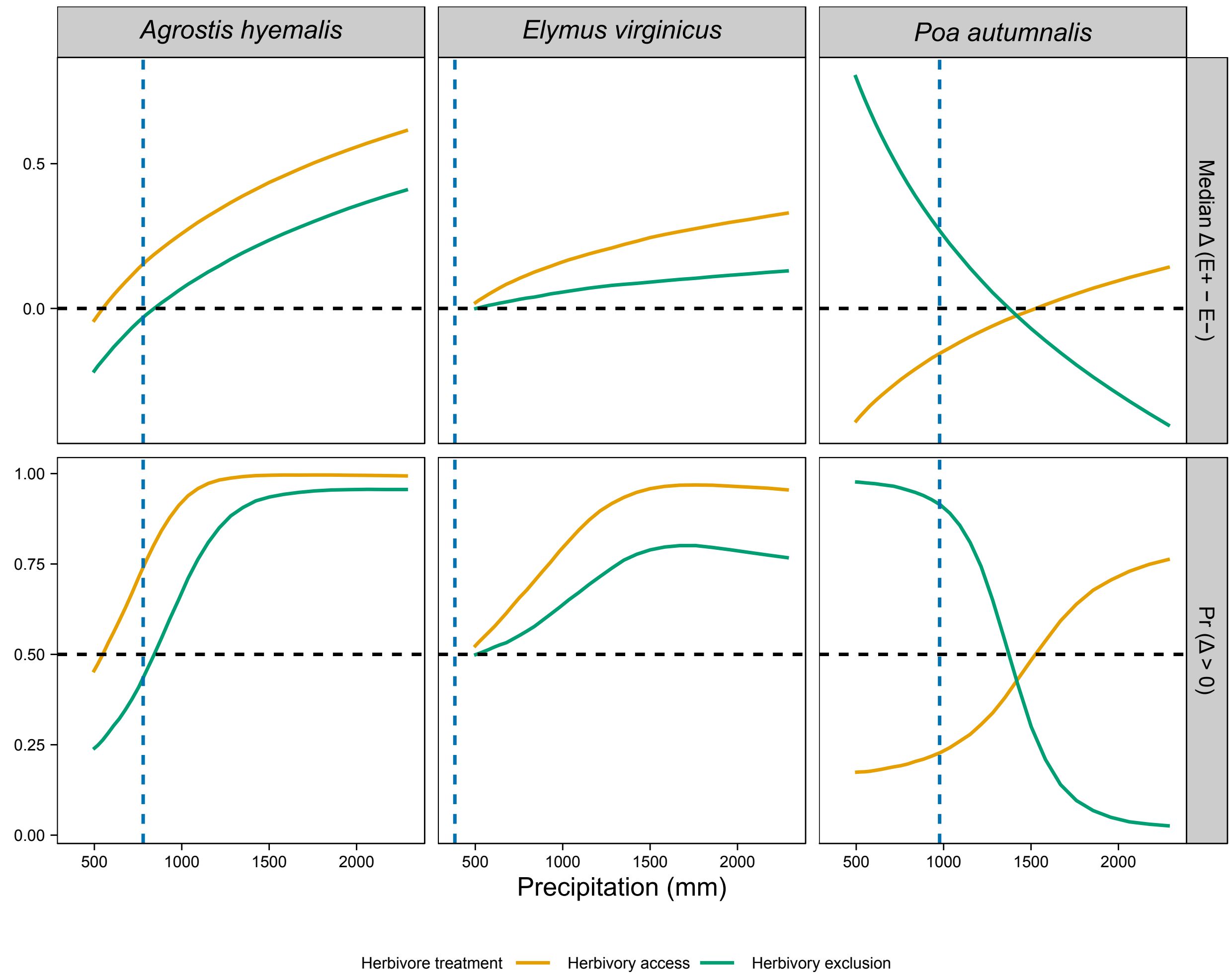

Fig S6: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^{+}) and endophyte-absent (E^{-}) plants (Median Δ, top row) and the posterior probability that this difference is positive ($\Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $\Pr(\Delta > 0) = 0.5$ (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

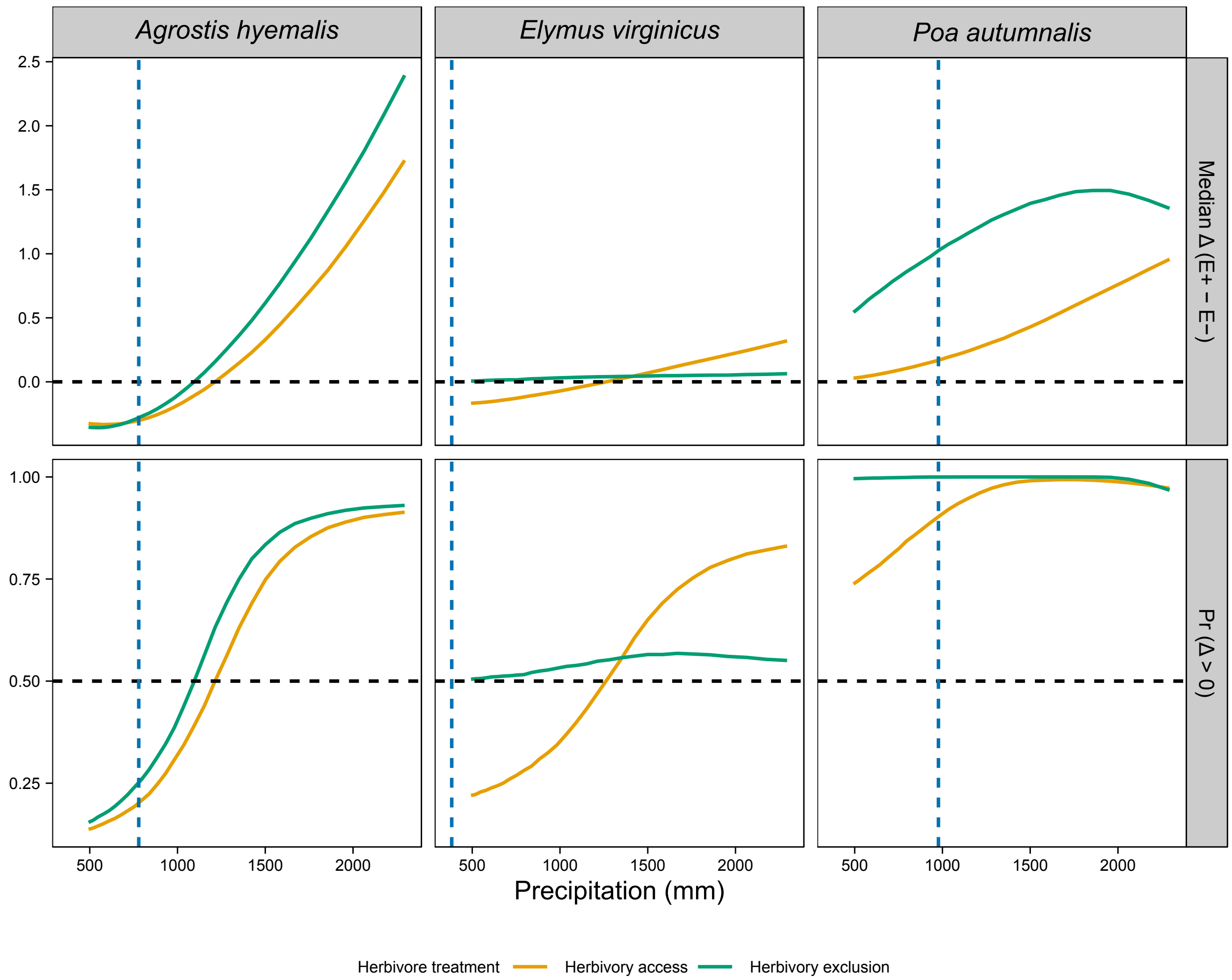

Fig S7: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference Δ ($E^+ - E^-$) and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

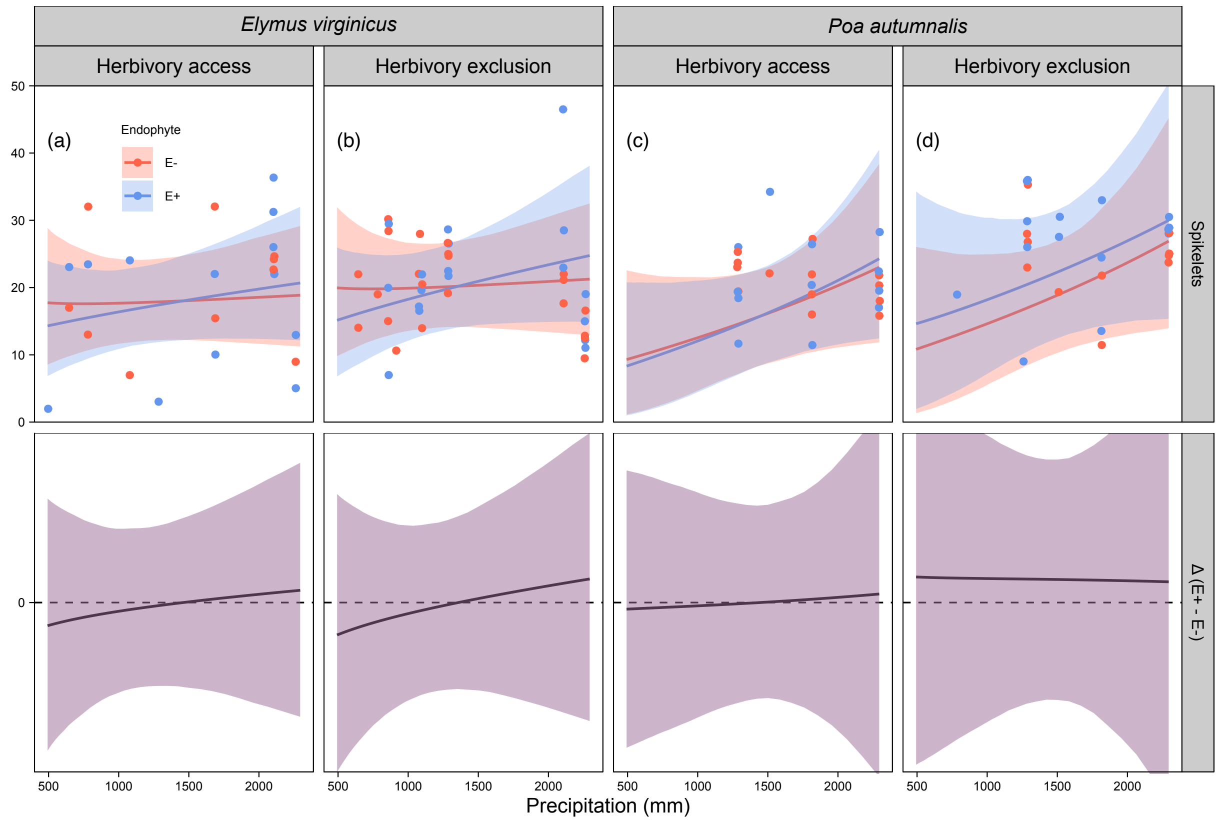

Fig S8: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

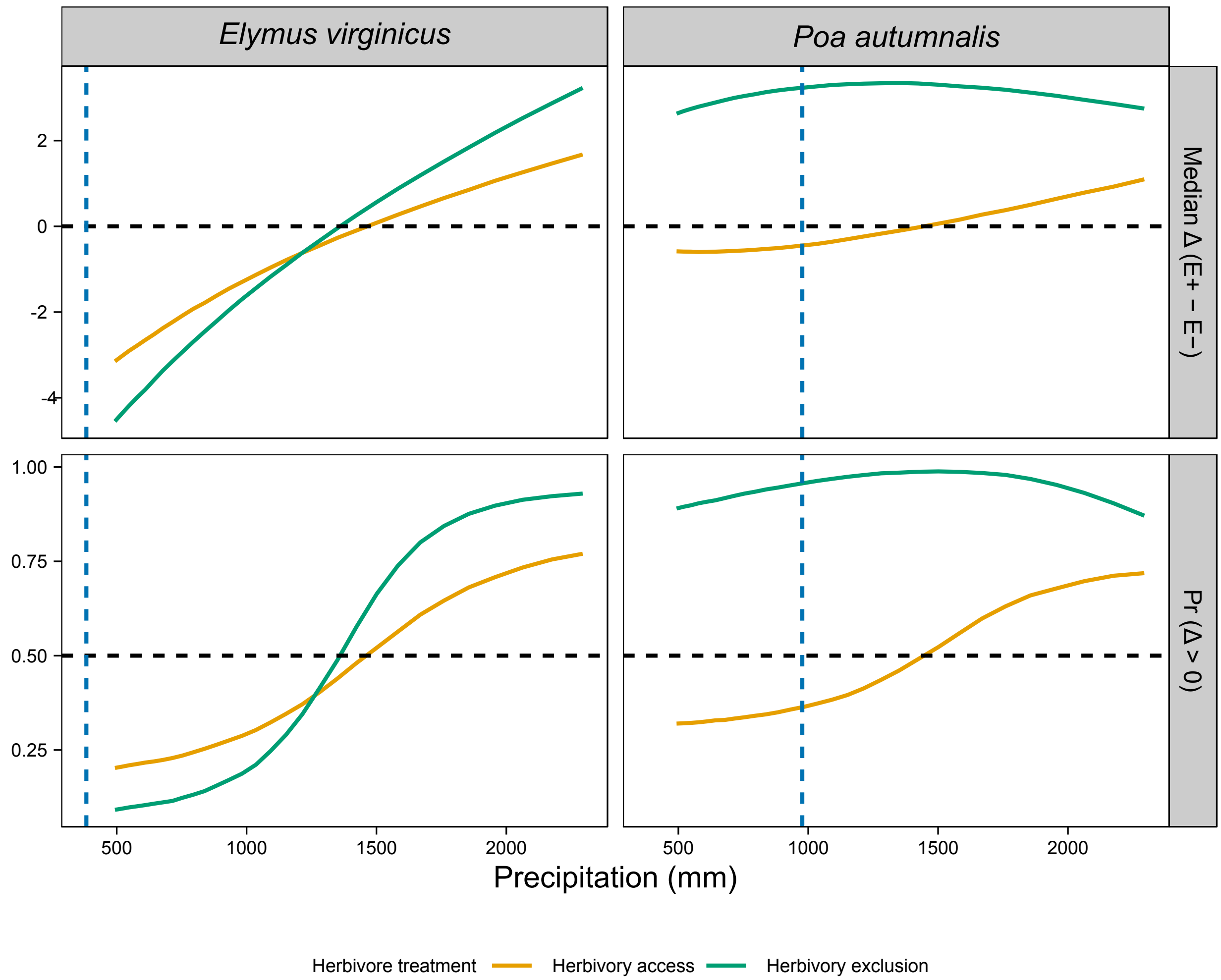

Fig S9: Effects of precipitation and endophyte presence on spikelet (reproductive output) for three cool-season grass species. Median Δ ($E^{+} - E^{-}$) and posterior probability $\Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $\Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.